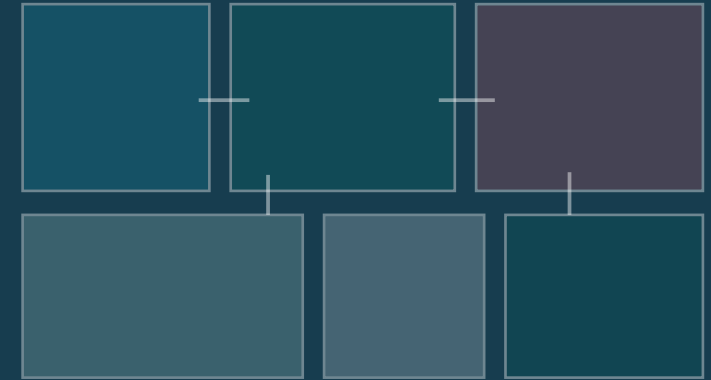


STAKEHOLDER QUESTION CATALOGUE 29

Space-planning Teams

80 planning decisions connecting current owner-controlled records, proportionate proposed evidence, explicit analytical operations, uncertainty, consultation and accountable boundaries.



Evidence for planning - not telemetry as planning authority

Use the catalogue to reconcile the current estate, expose evidence gaps, compare bounded scenarios and prepare decisions. Current source owners, measured surveys, consultation and competent professionals always take precedence.

80 QUESTIONS

SP-001 to SP-080, exactly ten in each of eight categories.

SIX FIELDS

Question, proposed evidence, authoritative data, analysis, boundary and value.

HISTORICAL PLAN

The May 2021 floor plan is spatial context only, never current operational truth.

ACCOUNTABLE USE

The service does not allocate space, certify, authorise or control.

1. Scope and accountable use

Evidence-aware space planning without transferring authority

This catalogue supports current-state reconciliation, evidence design, option comparison and decision preparation. It does not replace current data owners, competent professionals, affected-user consultation or institutional governance.

1

Decision support, not allocation

The service may look up, observe, calculate, estimate, forecast, compare, diagnose and recommend next work. It cannot allocate or withdraw space, approve a layout, change a booking, grant access or make an estate decision.

2

Current owners define facts

Space, booking, capacity, access, asset, service, safety, finance and project owners govern their domains. Analytics may expose a conflict but never silently overwrite an owner-issued status.

3

Historical floor-plan limit

The May 2021 Abacws floor plan is historical spatial reference only. It cannot establish current use, ownership, access, services, hazards, capacity, equipment, routes or closures.

4

Proposed until independently verified

Every sensor, integration, mapping and analytic remains proposed until permissioned, mapped to the exact scope, commissioned or validated and confirmed healthy for the decision interval.

5

Exact scope and time

A room, route, meter, asset or facility claim requires current verified mapping to that exact boundary and explicit effective, observation and retrieval times. Adjacent or shared evidence stays labelled as context.

6

Privacy and content minimisation

Use anonymous aggregates and numerical conditions only where proportionate. No identities, badge histories, device trails, speech, images, screens, communications, research content or network-content inspection.

7

Inclusion requires people and professionals

Use only functional requirements people choose to state. Do not infer protected traits or certify suitability. Consultation, measured surveys and competent accessibility, fire and safety input remain essential.

8

Honest safe failure

Every answer states its operation, interval, minimum evidence, uncertainty and authority. Missing, stale, conflicting, unauthorised or spatially invalid evidence produces a qualified answer, owner deferral or not assessable.

2. How to read every record

Six fields connect a planning question to evidence, method, uncertainty and authority

SPACE-PLANNING TEAMS QUESTION A natural decision question used to reconcile a fact, define an evidence gap, compare feasible options, prepare consultation or build an accountable decision pack.	SENSORS AND TELEMETRY Only physically plausible, privacy-preserving proposed evidence. Each record states modality or non-applicability, location, coverage, timing, aggregation and a claim-specific quality gate.	AUTHORITATIVE SUPPORTING DATA Current owner-controlled records with domain precedence, effective, event or observation and retrieval times, permissions, provenance, staleness and unresolved conflict handling.
ANALYSIS REQUIRED The operation type, decision interval, minimum evidence, method, uncertainty and safe failure. Lookup, observation, calculation, estimate, forecast, comparison, diagnosis and recommendation remain distinct.	ANSWER BOUNDARY What the service must not infer, disclose, promise, certify, authorise, allocate or control, and when it must defer to an owner or competent professional.	WHY THIS IS USEFUL The concrete action supported: correct a record, commission a survey, prepare consultation, identify a blocker, compare a trade-off or assemble a decision pack.

Operation taxonomy

LOOKUP retrieve a governed fact	OBSERVE summarise a measured state	CALCULATE apply a defined formula	ESTIMATE quantify with error	FORECAST project a future range	COMPARE show option differences	DIAGNOSE rank plausible contributors	RECOMMEND suggest bounded next work
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Safe answer states

SUPPORTED critical evidence is current, permissioned and in scope	QUALIFIED only a bounded subset is supported	CLARIFICATION ask the smallest question needed	OWNER DEFERRAL route authority to the named role	NOT ASSESSABLE critical evidence, mapping, permission or competence is absent
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3. Evidence, time and permission model

Use the least intrusive evidence that can support the exact claim and decision interval

A

OWNER-RECORD LED

Current owner records, controlled drawings, permissions, schedules, surveys and competent determinations. Telemetry is unnecessary or only corroborating context.

B

VERIFIED OBSERVATION

Anonymous aggregate telemetry or a measured survey is needed. Exact mapping, calibration or validation, clock, coverage, uptime, gaps, drift and lineage must pass the claim-specific gate.

C

GOVERNED SCENARIO

Forecasting, simulation, optimisation or recommendation uses approved assumptions, feasible options, uncertainty ranges, consultation and accountable owner decisions.

Claim-specific technical gate

1

Modality and placement

Name the sensor, integration or survey method and its physical or logical location. Record non-applicability instead of inventing sensing.

2

Coverage and aggregation

Map exact rooms, zones, entrances, routes, assets and service boundaries. State cadence, aggregation, exclusions and representative period.

3

Calibration and time

Retain calibration or validation, units and state semantics. Keep effective, event or observation and retrieval times explicit; align clocks to the analysis interval.

4

Health, gaps and drift

State expected refresh, uptime, gaps, saturation or occlusion, resets, drift, relocation, maintenance and stale-state timeout.

5

Provenance and permissions

Preserve owner, record or device ID, version, firmware or method, processing steps, permission scope, retention and disclosure controls.

6

Uncertainty and safe failure

Report measurement or scenario ranges. Never silently fill a critical gap, infer cause from association or select an option with unresolved hard constraints.

SOURCE PRECEDENCE

Current authoritative owners and competent professionals always take precedence. Equal-authority conflicts remain visible; the service qualifies, defers or returns not assessable rather than choosing silently.

1. Space inventory, ownership and current use

Questions 1-2 of 10

Category 1 of 8

SP-001 to SP-002 | 10 questions | Resolve identity, boundary, purpose, accountability and dependencies before any space enters a plan.

SP-001

INVENTORY BASELINE

OUTPUT: Reconciled baseline

EVIDENCE: Owner records

SPACE-PLANNING TEAMS QUESTION

Which rooms, zones, shared areas and support spaces belong in the current planning baseline, and which records still need correction or owner confirmation?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: room and asset registry heartbeats that confirm whether mapped identifiers are active, without observing people or content. Spatial coverage: every room, shared zone and non-occupiable support area included in the current space register; no adjacent-space substitution. Sampling and aggregation: event-driven metadata with a daily health summary for the current planning snapshot. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: the institutional space-management owner, Estates information owner and accountable local space owners. Records: the current room inventory, area schedule, official identifiers, occupiable or support-space classification, ownership fields and approved status flags. Time basis: the named current planning snapshot and any effective future changes already authorised. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current space and Estates owners, controlled drawings and authorised change records govern identity, purpose, rights, boundaries and services; lower-authority evidence may only flag a discrepancy. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: authorised lookup and identifier reconciliation. Decision interval: the current planning snapshot. Minimum evidence: a complete current inventory extract plus cross-system room identifiers and accountable owners. Method: join records by verified identifiers and spatial relationships, classify each space as confirmed, conflicting, missing or retired, and report portfolio coverage. Uncertainty: unregistered partitions, delayed updates, duplicate room labels and spaces omitted from one source. Safe failure: return the supported subset, expose each gap and direct the missing record to the named owner; otherwise return not assessable.

ANSWER BOUNDARY

Do not create a room record, declare that an omitted area does not exist, or use a drawing label as current status.

WHY THIS IS USEFUL

Creates a defensible inventory baseline before utilisation, capacity or change scenarios are calculated.

SP-002

DECISION RIGHTS

OUTPUT: Accountability map

EVIDENCE: Owner records

SPACE-PLANNING TEAMS QUESTION

For this space and proposed change, who owns each record, operates or funds the space, must be consulted, and can authorise the decision?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: role-based owner-directory status and controlled entitlement metadata; no badge histories or named occupant locations. Spatial coverage: each named room or zone and the decision right being queried, not a whole-person access profile. Sampling and aggregation: current-state lookup with effective-from and expiry dates; no behavioural aggregation. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: the institutional space authority, building owner, School or unit space leads, security/access-control owner and service owners. Records: delegations, licences or leases, occupation agreements, local stewardship assignments, booking authority, access authority and change-approval routes. Time basis: the requested decision and proposed effective date; future delegations are shown only when formally recorded. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current space and Estates owners, controlled drawings and authorised change records govern identity, purpose, rights, boundaries and services; lower-authority evidence may only flag a discrepancy. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup. Technique: authority and responsibility lookup. Decision interval: the current decision window. Minimum evidence: a verified space identifier, the type of decision and current role-based authority records. Method: resolve the room to accountable, consulted, informed and operational roles; distinguish use permission from approval authority. Uncertainty: temporary delegations, overlapping ownership, vacant roles and conflicting directories. Safe failure: show the conflict and refer the decision to the institutional space authority; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer authority from who appears to use a room, disclose unnecessary personal contact details, or treat technical access as approval power.

WHY THIS IS USEFUL

Lets planners contact the correct accountable role and avoid scenarios that assume authority they do not have.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

1. Space inventory, ownership and current use

Questions 3-4 of 10

Category 1 of 8

SP-003 to SP-004 | 10 questions | Resolve identity, boundary, purpose, accountability and dependencies before any space enters a plan.

SP-003

IDENTIFIER CROSSWALK

OUTPUT: Verified crosswalk

EVIDENCE: Records + metadata

SPACE-PLANNING TEAMS QUESTION

Can every candidate space be matched unambiguously across the space register, drawings, booking, access, assets, meters, AV, network and BMS records?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: device and gateway metadata, meter-point identities and service-zone tags; no environmental measurements are needed. Spatial coverage: the named room plus directly served assets, access points, meters and control zones; upstream context stays separately labelled. Sampling and aggregation: current metadata snapshot with event times for recent remapping or device relocation. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space information, room booking, security/access, BMS, utility-meter, network, audiovisual and asset- register owners. Records: current identifier dictionaries, room-to-zone maps, served-area definitions, asset locations, meter hierarchies and approved topology changes. Time basis: the decision interval and the latest effective mapping for every included system. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current space and Estates owners, controlled drawings and authorised change records govern identity, purpose, rights, boundaries and services; lower-authority evidence may only flag a discrepancy. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: entity resolution and mapping-confidence classification. Decision interval: the current evidence interval. Minimum evidence: the authoritative room identifier plus current dictionaries from every system required by the intended decision. Method: construct a provenance-preserving crosswalk, test one-to-one and one-to-many relationships, flag unmapped or ambiguous links and calculate coverage. Uncertainty: shared meters, multi-room HVAC zones, moved devices, renamed rooms and unsynchronised source updates. Safe failure: exclude ambiguous links from room-level claims and state which owner must resolve each mapping; otherwise return not assessable.

ANSWER BOUNDARY

Do not relabel zone-level or shared-service evidence as a direct room measurement, and do not expose restricted access or infrastructure topology beyond permission.

WHY THIS IS USEFUL

Prevents apparently precise analyses from joining evidence that belongs to different rooms, assets or service boundaries.

SP-004

APPROVED PURPOSE AND CONDITIONS

OUTPUT: Purpose schedule

EVIDENCE: Owner records

SPACE-PLANNING TEAMS QUESTION

What purpose and activities are currently authorised for each space on the decision date, and what restrictions, expiry dates or conditions apply?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional non-personal room- status heartbeats for opening, closure or service availability; sensing cannot establish purpose. Spatial coverage: the exact named room and each separately governed sub-area. Sampling and aggregation: event-driven status for the requested date, with a current health timestamp. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: the institutional space owner, local space owner, room-booking owner, accessibility owner, facilities service owners and security. Records: current room purpose, bookability, eligible activities, opening and access rules, service level, restrictions, approved layouts and authorised future changes. Time basis: now or the user- specified future interval, using records effective for that interval. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current space and Estates owners, controlled drawings and authorised change records govern identity, purpose, rights, boundaries and services; lower-authority evidence may only flag a discrepancy. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup. Technique: authorised classification lookup. Decision interval: the requested present or future interval. Minimum evidence: a verified room identifier and current purpose, eligibility, access and service records. Method: separate official purpose from bookability, temporary status and local custom; list confirmed constraints and unresolved fields. Uncertainty: purpose changes awaiting publication, temporary closures and differing local terminology. Safe failure: give only current confirmed fields and route unknowns to the accountable owner; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer purpose from furniture, noise, occupancy or power use, and do not present temporary vacancy as permission to use or repurpose the room.

WHY THIS IS USEFUL

Gives every later capacity, suitability and utilisation calculation the correct authoritative starting point.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

Stakeholder Question Catalogue 29 | Category 1: Space inventory, ownership and current use

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1. Space inventory, ownership and current use

Questions 5-6 of 10
Category 1 of 8

SP-005 to SP-006 | 10 questions | Resolve identity, boundary, purpose, accountability and dependencies before any space enters a plan.

SP-005

PURPOSE-USE OWNER CHECK

OUTPUT: Owner-check list

EVIDENCE: Aggregate observation

SPACE-PLANNING TEAMS QUESTION

Which spaces show a sustained difference between approved purpose or service assumptions and supported anonymous aggregate use that should be checked with the owner?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous doorway counts or ceiling-zone totals, use-duration bands, non-content plug or lighting load and numerical acoustic statistics only as separately labelled context, never as direct occupancy measures. Spatial coverage: the named room or shared zone only, with complete eligible entrances and no device identifiers or movement histories. Sampling and aggregation: one-minute observations aggregated to 15-minute bands over at least six representative weeks, excluding closures and exceptional events. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: the current space-purpose owner, room-booking and timetable owners, facilities service owners and sensor-data steward. Records: official purpose, eligible activities, approved capacity and layout, bookings, cancellations, events, opening, closures and known temporary uses. Time basis: the agreed representative period with effective, observed and retrieval times preserved. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current space and Estates owners, controlled drawings and authorised change records govern identity, purpose, rights, boundaries and services; lower-authority evidence may only flag a discrepancy. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation and comparison. Technique: stratified comparison. Decision interval: the agreed representative period. Minimum evidence: current purpose and eligibility plus valid room-level aggregate use across enough comparable days. Method: compare distributions of use intensity, duration, sound and service demand with owner-stated service assumptions; identify mismatches without reclassifying the room. Uncertainty: counting error, unscheduled authorised use, shared signals, seasonal demand and unrecorded temporary changes. Safe failure: report association and evidence gaps, not a purpose judgement, and defer classification to the owner; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer the activity, identity, quality or value of use, or treat a statistical mismatch as permission to change purpose.

WHY THIS IS USEFUL

Identifies rooms where owner records, service assumptions or evidence coverage may need correction before planning decisions.

SP-006

AVAILABILITY AND COMMITMENTS

OUTPUT: Availability status

EVIDENCE: Records + status

SPACE-PLANNING TEAMS QUESTION

Which apparently unused spaces are actually closed, restricted, project-held, committed, awaiting works or subject to an approved future change?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous room-level presence totals and authorised door or service-status heartbeats; no badge histories or occupant identification. Spatial coverage: only spaces with verified room-level coverage and the affected access or service boundary. Sampling and aggregation: one-minute presence aggregated to 15-minute intervals; event-driven closure and asset status for the selected period. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space and room owners, booking and timetable owners, facilities/maintenance, project, security and access-control owners. Records: current bookings and holds, closures, restricted-use classifications, project possession, maintenance windows, decant reservations and authorised future commitments. Time basis: the candidate planning period, including records that become effective within it. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current space and Estates owners, controlled drawings and authorised change records govern identity, purpose, rights, boundaries and services; lower-authority evidence may only flag a discrepancy. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, observation and comparison. Technique: availability-status classification. Decision interval: the candidate planning period. Minimum evidence: current owner-issued restrictions and commitments plus valid aggregate evidence where available. Method: separate observed low use from booked, held, restricted, closed, under-work, unobserved and genuinely uncommitted states. Uncertainty: late cancellations, hidden dependencies, sensor blind spots and future records not yet integrated. Safe failure: never label a space available unless the owner system confirms eligibility; otherwise show the governing restriction or unknown; otherwise return not assessable.

ANSWER BOUNDARY

Do not equate low detections with vacancy, expose sensitive project or security details, or suggest occupation of a held, restricted or closed space.

WHY THIS IS USEFUL

Stops planners from treating apparently quiet rooms as supply when authoritative commitments make them unavailable.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

1. Space inventory, ownership and current use

Questions 7-8 of 10

Category 1 of 8

SP-007 to SP-008 | 10 questions | Resolve identity, boundary, purpose, accountability and dependencies before any space enters a plan.

SP-007

SHARING AND BOOKING RULES

OUTPUT: Use-rights matrix

EVIDENCE: Owner records

SPACE-PLANNING TEAMS QUESTION

For the requested period, which spaces are exclusive, pooled, shared, centrally bookable, locally managed, open-access, licensed or otherwise restricted?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional room-status and booking-system heartbeats that show source health, not who is using the space. Spatial coverage: each named space and any separately governed zone within it. Sampling and aggregation: current-state lookup plus effective dates for future rule changes. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: the institutional space authority, room-booking owner, local space owner, security/access owner and relevant service owners. Records: occupation category, sharing agreement, booking channel, priority rules, access conditions, opening hours, exceptions and approved temporary arrangements. Time basis: the date and activity being planned, using rules effective at that time. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current space and Estates owners, controlled drawings and authorised change records govern identity, purpose, rights, boundaries and services; lower-authority evidence may only flag a discrepancy. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup. Technique: authorised rule and eligibility lookup. Decision interval: the requested use interval. Minimum evidence: verified space identity, proposed activity and current ownership, booking and access rules. Method: classify the space by management and use regime; list the applicable request route and any unresolved conflicts. Uncertainty: overlapping regimes, temporary local arrangements and rules that differ by time or activity. Safe failure: withhold an eligibility conclusion and refer to the accountable owner when the governing regime is unclear; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer openness from an unlocked door or observed use, bypass the booking route, or disclose restricted access conditions beyond the planner's permission.

WHY THIS IS USEFUL

Shows which supply can legitimately enter a planning scenario and which requires a separate owner decision.

SP-008

BOUNDARIES AND MEASURED AREA

OUTPUT: Survey action list

EVIDENCE: Records + measured survey

SPACE-PLANNING TEAMS QUESTION

Where do current drawings, room-register boundaries, subdivisions, mergers or measured areas disagree, and what survey or change control is needed?

SENSORS AND TELEMETRY

The core answer is record- and survey-led. Proposed evidence: a controlled measured survey or an unoccupied laser scan of the disputed room boundary, partitions, doors and fixed obstructions; no occupancy capture. Spatial coverage: the exact disputed area, including both sides of shared partitions and any affected support space. Timing and aggregation: one dated survey for the decision baseline and a repeat after authorised works; retain raw measurements and the approved derived area schedule. Quality gate: retain survey method, instrument calibration or validation, survey date, coordinate reference, tolerance, full boundary or route coverage, obscured areas, revision and competent provenance. A survey is a dated observation, not approval. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: the Estates information and space-management owners, competent measured-survey owner, project/change-control owner and accountable local space owner. Records: current controlled drawings, room register, area schedule, approved partition or merger changes and survey deliverables. Time basis: the decision date and the effective date of every physical change. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current space and Estates owners, controlled drawings and authorised change records govern identity, purpose, rights, boundaries and services; lower-authority evidence may only flag a discrepancy. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, comparison and recommendation. Decision interval: the current baseline and any authorised effective change. Minimum evidence: controlled drawing and register versions, verified room identifiers, declared area definitions and either a current measured survey or formal change record. Method: compare boundaries and areas by version; classify each discrepancy as drawing lag, register lag, unresolved physical change or measurement-method difference; name the owner action. Uncertainty: inaccessible areas, differing area standards and unrecorded partitions. Safe failure: show the conflict and commission survey or change-control resolution; do not select a boundary or area value.

ANSWER BOUNDARY

Do not choose a convenient boundary, convert gross and net area definitions without an approved rule, or treat a historical drawing as proof of current construction. Refer unresolved geometry to the Estates information owner and competent surveyor.

WHY THIS IS USEFUL

Prevents area, capacity and scenario calculations from being built on the wrong room boundary or measurement basis.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

Stakeholder Question Catalogue 29 | Category 1: Space inventory, ownership and current use

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1. Space inventory, ownership and current use

Questions 9-10 of 10

Category 1 of 8

SP-009 to SP-010 | 10 questions | Resolve identity, boundary, purpose, accountability and dependencies before any space enters a plan.

SP-009

SPACE-SERVICE DEPENDENCIES

OUTPUT: Dependency map

EVIDENCE: Records + asset metadata

SPACE-PLANNING TEAMS QUESTION

Which current rooms depend on which HVAC zones, meters, power, water, network, AV, storage, loading or specialist services, and where is the mapping uncertain?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: asset heartbeats, controller identifiers, meter-channel metadata and content-free equipment status where authorised. Spatial coverage: the named room, each directly serving asset and any shared upstream zone shown separately. Sampling and aggregation: current metadata with last-seen, commissioning, relocation and maintenance event times. Quality gate: verify asset identity, room or service mapping and state semantics through commissioning or stocktake; retain event time, stale-state timeout, expected heartbeat, uptime, gaps, relocations, duplicates, firmware and source lineage. A heartbeat is not proof of suitability. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: asset-register, BMS, utilities, IT/network, AV, facilities, laboratory or specialist-equipment and space- information owners. Records: current asset inventory, room and served-zone links, meter hierarchy, isolation boundaries, service capacity, ownership and approved changes. Time basis: the intended planning horizon and any future service changes effective within it. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current space and Estates owners, controlled drawings and authorised change records govern identity, purpose, rights, boundaries and services; lower-authority evidence may only flag a discrepancy. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: mapping-completeness assessment. Decision interval: the current and stated future planning horizon. Minimum evidence: a verified room list and current service/asset mappings from every relevant owner. Method: build a room-service graph, score confirmed, shared, ambiguous and missing links, and separate status from capacity or suitability. Uncertainty: shared systems, undocumented local equipment, moved assets and stale controller names. Safe failure: exclude unsupported services from suitability claims and identify the exact mapping evidence required; otherwise return not assessable.

ANSWER BOUNDARY

Do not expose sensitive infrastructure details, infer capability from a powered device alone, or claim that mapped equipment is available, maintained or fit for a proposed activity.

WHY THIS IS USEFUL

Reveals whether a candidate space can be assessed at all for service-intensive teaching, research, workplace or event uses.

SP-010

PERMISSIONED PLANNING EXTRACT

OUTPUT: Shareable baseline

EVIDENCE: Permissioned records

SPACE-PLANNING TEAMS QUESTION

What current space-inventory view can this planning group use, and which fields must be suppressed, aggregated, caveated or confirmed by an owner?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional portfolio totals may use only anonymous aggregates that meet disclosure thresholds. Spatial coverage: the authorised portfolio scope and audience, excluding security-sensitive zones and small sensitive groupings. Sampling and aggregation: a dated snapshot with source retrieval times and a stated validity period. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: the institutional space authority, information-governance owner, security, accessibility and each contributing data owner. Records: approved inventory fields, ownership roles, purpose, capacity, availability status, aggregate utilisation, planned changes and disclosure classifications. Time basis: the publication date and intended audience. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current space and Estates owners, controlled drawings and authorised change records govern identity, purpose, rights, boundaries and services; lower-authority evidence may only flag a discrepancy. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup. Technique: permission-aware extraction and minimisation. Decision interval: the stated communication window. Minimum evidence: current source permissions, audience classification and sufficiently aggregated evidence. Method: select the minimum fields needed, suppress restricted or small-cell details, attach provenance and separate confirmed facts from estimates. Uncertainty: changing permissions, mixed-security portfolios and totals built from incomplete coverage. Safe failure: publish only the permitted supported subset and name omitted categories without revealing their content; otherwise return not assessable.

ANSWER BOUNDARY

Do not disclose named occupants, access-control detail, sensitive research locations, personal schedules or precise restricted infrastructure; do not present estimates as official allocations.

WHY THIS IS USEFUL

Provides a trustworthy common evidence pack for discussion while preserving privacy, security and source ownership.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

Stakeholder Question Catalogue 29 | Category 1: Space inventory, ownership and current use

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2. Occupancy, utilisation and evidence quality

SP-011 to SP-012 | 10 questions | Use representative anonymous evidence, fair denominators and explicit quality grades rather than a single utilisation score.

Questions 1-2 of 10

Category 2 of 8

SP-011

REPRESENTATIVE USE PATTERN

OUTPUT: Representative profile

EVIDENCE: Aggregate counts

SPACE-PLANNING TEAMS QUESTION

For each space, what occupied and unoccupied time pattern is sufficiently stable and representative to use in planning?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous doorway counts or ceiling-zone totals, with non-content lighting or plug-load signals only as separately labelled corroboration. Spatial coverage: each eligible room and shared zone with complete entrances; no cross-room device tracing or individual movement paths. Sampling and aggregation: one-minute observations aggregated to 15-minute bands over a full teaching cycle where possible, with term, vacation and event periods separated. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, booking, timetable, event, opening-hours, facilities and sensor-data owners. Records: current room purpose, bookings, cancellations, scheduled activities, approved events, closures, service windows and academic calendar. Time basis: the selected cycle, preserving the effective schedule and observed telemetry times. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Owner records govern eligibility, opening, bookings, closures and approved capacity; telemetry describes anonymous aggregate observations only and cannot overwrite those facts. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation, comparison and estimate. Technique: time-series segmentation and pattern classification. Decision interval: the selected term, year or comparable operating cycle. Minimum evidence: valid room-level aggregates over enough representative weeks plus current schedule and closure context. Method: separate normal, intermittent, seasonal and exceptional intervals; calculate recurrence, duration and variability with coverage-weighted confidence. Uncertainty: partial cycles, timetable changes, sensor outages and rare authorised activity. Safe failure: label insufficiently observed spaces as unknown rather than forcing a pattern class; otherwise return not assessable.

ANSWER BOUNDARY

Do not identify users, infer the nature or value of an activity, or describe irregular use as misuse or poor performance.

WHY THIS IS USEFUL

Helps planners distinguish genuinely recurring demand from occasional peaks before changing space supply.

SP-012

PEAK USE AGAINST CAPACITY

OUTPUT: Peak-risk profile

EVIDENCE: Counts + capacity

SPACE-PLANNING TEAMS QUESTION

During representative operating periods, when and how often does anonymous aggregate occupancy approach the currently approved capacity?

SENSORS AND TELEMETRY

Proposed modality and location: commissioned anonymous entrance counts, bidirectional doorway counters or ceiling-zone totals appropriate to the room geometry. Spatial coverage: the named room with all eligible entrances and known internal subdivisions; adjacent corridor counts are context only. Sampling and aggregation: one-minute counts with five- or 15-minute occupancy bands for the decision period; retain native event times. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: capacity and layout owners, fire-safety and accessibility owners, room-booking/timetable owners and sensor-data steward. Records: current approved capacity for the applicable layout, layout validity, bookings, closures, event setup and any temporary capacity restrictions. Time basis: the selected operating periods, using the capacity effective at each time. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Owner records govern eligibility, opening, bookings, closures and approved capacity; telemetry describes anonymous aggregate observations only and cannot overwrite those facts. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation, calculation and comparison. Technique: occupancy-band calculation and threshold comparison. Decision interval: the stated sessions, days or representative period. Minimum evidence: approved capacity by valid layout and aggregate counts with quantified error for the whole room. Method: calculate conservative occupancy bands, compare their upper bounds with agreed planning thresholds and disclose excluded intervals. Uncertainty: counter error, door propping, multi-entrance imbalance, layout changes and people moving between sub-zones. Safe failure: do not make a capacity claim when layout or count coverage is uncertain; provide source status and refer high-consequence concerns; otherwise return not assessable.

ANSWER BOUNDARY

Do not claim a statutory breach, certify safe occupancy, identify attendance, or use a sensor estimate to admit, exclude or relocate people.

WHY THIS IS USEFUL

Shows where demand margins may be thin enough to require a formal capacity or scheduling assessment.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

2. Occupancy, utilisation and evidence quality

SP-013 to SP-014 | 10 questions | Use representative anonymous evidence, fair denominators and explicit quality grades rather than a single utilisation score.

Questions 3-4 of 10

Category 2 of 8

SP-013

PERSISTENT LOW USE

OUTPUT: Investigation shortlist

EVIDENCE: Counts + owner records

SPACE-PLANNING TEAMS QUESTION

Which comparable spaces remain consistently low-use after removing closures, restricted functions, booked commitments, exceptional weeks and unsupported intervals?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous room-level counts or presence totals, with optional non-content lighting or plug-load indicators used only as corroboration. Spatial coverage: all compared rooms must have equivalent room-level coverage and complete eligible entrances; no corridor proxy substitution. Sampling and aggregation: one-minute observations aggregated to 15-minute occupied bands across at least six comparable weeks per season or term. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space and room owners, booking/timetable owners, opening and closure owners, facilities and sensor-data owners. Records: current purpose, capacity, eligible hours, bookings, cancellations, special uses, closures, maintenance, future commitments and service restrictions. Time basis: the agreed representative period with exceptional intervals identified. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Owner records govern eligibility, opening, bookings, closures and approved capacity; telemetry describes anonymous aggregate observations only and cannot overwrite those facts. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation, calculation and comparison. Technique: stratified utilisation calculation and comparison. Decision interval: the agreed representative period. Minimum evidence: current eligibility denominators, valid comparable telemetry and authoritative context for exclusions. Method: calculate occupied eligible hours and use-intensity distributions by room class; show confidence intervals, denominator choices and sensitivity to missing data. Uncertainty: unrecorded authorised use, differing room functions, seasonal demand and incomplete monitoring. Safe failure: classify only evidence-supported low use and keep every space with weak coverage in an unknown group; otherwise return not assessable.

ANSWER BOUNDARY

Do not call a room unnecessary, surplus or available, infer the importance of its activities, or trigger allocation change from low aggregate use.

WHY THIS IS USEFUL

Creates a defensible shortlist for owner consultation and deeper suitability analysis rather than a simplistic vacancy list.

SP-014

CROWDING AND OVERFLOW

OUTPUT: Hotspot schedule

EVIDENCE: Count-only route evidence

SPACE-PLANNING TEAMS QUESTION

Where and when do repeated crowding, overflow or transition peaks affect room entrances, waiting areas or route segments?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous doorway counts, ceiling-zone density totals and route-segment counters; optional numerical sound levels without recording speech. Spatial coverage: named rooms, waiting areas and route segments with continuous coverage at decision points; no linkage of detections across sensors. Sampling and aggregation: one-minute aggregates around scheduled transitions, summarised by comparable day and event type. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, timetable, booking, event, circulation, security, accessibility and facilities owners. Records: current room assignments, transition times, route graph, door/lift status, waiting-area rules, closures and approved events. Time basis: the selected transitions and comparable operating weeks. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Owner records govern eligibility, opening, bookings, closures and approved capacity; telemetry describes anonymous aggregate observations only and cannot overwrite those facts. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation and comparison. Technique: spatial-temporal hotspot comparison. Decision interval: the selected transitions and representative period. Minimum evidence: valid aggregate coverage of the room exits and affected route segments plus current schedule and route context. Method: calculate duration above agreed planning bands, distinguish queues from through-flow where evidence permits and compare matched transitions. Uncertainty: counter overlap, people pausing for unrelated reasons, event anomalies and route diversions. Safe failure: report only the covered hotspot and time window; do not extrapolate to unobserved routes; otherwise return not assessable.

ANSWER BOUNDARY

Do not track journeys, infer which cohort or event caused the pattern, certify crowd safety, or direct operational crowd control.

WHY THIS IS USEFUL

Supports targeted investigation of layouts, transition timing, waiting space and route alternatives.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

2. Occupancy, utilisation and evidence quality

SP-015 to SP-016 | 10 questions | Use representative anonymous evidence, fair denominators and explicit quality grades rather than a single utilisation score.

Questions 5-6 of 10

Category 2 of 8

SP-015

BOOKING-USE MISMATCH

OUTPUT: Mismatch queue

EVIDENCE: Bookings + counts

SPACE-PLANNING TEAMS QUESTION

Which booked sessions lack supported aggregate use, and which unbooked-use signals need owner investigation rather than automatic classification?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous room-level entrance counts or presence totals; optional room service-state heartbeat to distinguish unavailable periods. Spatial coverage: the booked room only, with complete entrances and no association with invitees, devices or named courses. Sampling and aggregation: one-minute observations aggregated to session-level occupied bands, including setup and release windows. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: room-booking and timetable owners, event organisers, room/space owners, facilities and sensor-data owners. Records: current bookings, cancellations, releases, linked rooms, setup windows, approved room swaps, closures and service outages. Time basis: the selected term or booking sample, using the transaction state effective at each session. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Owner records govern eligibility, opening, bookings, closures and approved capacity; telemetry describes anonymous aggregate observations only and cannot overwrite those facts. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, observation and comparison. Technique: booking-observation comparison. Decision interval: the selected sessions or term. Minimum evidence: authoritative booking transactions and valid aggregate room-level use for the same aligned intervals. Method: classify confirmed use, likely no-use, partial use, timing mismatch, room swap, closure and unknown; report count-error and source conflicts. Uncertainty: late room changes, setup activity, very small groups and incomplete entrance coverage. Safe failure: do not label a no-show when evidence cannot distinguish low use from sensor failure or authorised alternative use; otherwise return not assessable.

ANSWER BOUNDARY

Do not identify attendees, infer absence by named groups, release bookings automatically, penalise requesters or disclose private booking details.

WHY THIS IS USEFUL

Shows where booking rules, release arrangements or data integration need correction without turning sensors into attendance monitors.

SP-016

CONTEXT-SENSITIVE USE

OUTPUT: Context comparison

EVIDENCE: Records + observation

SPACE-PLANNING TEAMS QUESTION

How does aggregate use change by time band, day type, term phase, planned event or closure, and which apparent differences disappear after fair stratification?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous room and route counts, with optional numerical temperature, CO2, illuminance and sound indicators only as contextual covariates, not occupancy measures or causes. Spatial coverage: each compared room or zone with equivalent coverage; external weather from an authorised source rather than inferred indoors. Sampling and aggregation: one- or five-minute observations aggregated to 15-minute bands across matched days and event classes. Quality gate: document sensor height and representative placement, units and calibration or validation; align clocks within 60 seconds; report cadence, spatial variation, uptime, gaps, saturation, drift, maintenance and processing lineage. Readings describe sampled conditions only. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, timetable, booking, event, opening, weather, facilities and sensor-data owners. Records: academic calendar, authorised events, closures, current room purpose, eligible hours, weather observations and known service disruptions. Time basis: the selected historical period and the comparator definitions agreed by planners. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Owner records govern eligibility, opening, bookings, closures and approved capacity; telemetry describes anonymous aggregate observations only and cannot overwrite those facts. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation and comparison. Technique: multivariable stratified comparison. Decision interval: the selected historical period. Minimum evidence: sufficient matched observations across each context class and current authoritative labels for those classes. Method: compare use distributions while controlling for room type, eligible hours and scheduled demand; report associations and interaction ranges. Uncertainty: confounding between weather, calendar and events, sparse rare conditions and changing room purpose. Safe failure: omit unsupported factors, show sensitivity and avoid causal language; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer why individuals chose a space, infer health or comfort outcomes, or use contextual associations to profile groups.

WHY THIS IS USEFUL

Helps planners choose representative baselines and understand when a single annual average hides material variation.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

Stakeholder Question Catalogue 29 | Category 2: Occupancy, utilisation and evidence quality

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2. Occupancy, utilisation and evidence quality

SP-017 to SP-018 | 10 questions | Use representative anonymous evidence, fair denominators and explicit quality grades rather than a single utilisation score.

Questions 7-8 of 10

Category 2 of 8

SP-017DECISION-GRADE UTILISATION EVIDENCEOUTPUT: Evidence-grade mapEVIDENCE: Source-health evidence

SPACE-PLANNING TEAMS QUESTIONWhich proposed utilisation statements are supported, qualified or not assessable for the current planning decision?

SENSORS AND TELEMETRY

Proposed modality and location: sensor registry, commissioning evidence, calibration or validation results, clock status, uptime, gap, drift, relocation and maintenance metadata. Spatial coverage: every sensor and entrance needed for each proposed room, route or zone statement; indirect or shared points remain separately labelled. Sampling and aggregation: quality assessed for the exact requested interval, with native observations retained and analysis aggregation declared. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: sensor-data steward, space-information owner, Estates/IT platform owners and each decision-domain owner. Records: current sensor catalogue, room and asset mappings, commissioning records, validation samples, permissions, retention rules and source-health history. Time basis: the requested evidence interval and analytical purpose. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Owner records govern eligibility, opening, bookings, closures and approved capacity; telemetry describes anonymous aggregate observations only and cannot overwrite those facts. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: evidence-sufficiency classification. Decision interval: the exact interval and claim being assessed. Minimum evidence: a claim specification plus complete quality and provenance metadata for every required point. Method: test spatial validity, temporal completeness, error tolerance, comparator consistency and permission; classify direct, contextual, estimated or unsupported evidence. Uncertainty: unknown error under unusual conditions, silent relocation and quality metadata that is itself stale. Safe failure: remove invalid points, narrow the claim to supported scope or state not assessable; otherwise return not assessable.

ANSWER BOUNDARY

Do not hide quality gaps behind a single confidence score, fill missing periods with unsupported model output, or present contextual data as direct observation.

WHY THIS IS USEFUL

Prevents weak telemetry from becoming apparently authoritative utilisation evidence in high-impact space decisions.

SP-018EVIDENCE-GAP SAMPLING PLANOUTPUT: Proportionate evidence planEVIDENCE: Coverage metadata

SPACE-PLANNING TEAMS QUESTIONWhere are the planning-critical utilisation blind spots, and what is the least intrusive observation or survey needed to close each gap?

SENSORS AND TELEMETRY

Proposed modality and location: the proposed sensor catalogue, existing anonymous count coverage and non-personal source-health metadata; no new collection is assumed. Spatial coverage: all rooms and shared areas in scope, classified by whether direct, contextual or no eligible telemetry exists. Sampling and aggregation: coverage calculated for the intended planning periods, not merely current heartbeat status. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space-information, booking/timetable, sensor-data, privacy, security, accessibility and facilities owners. Records: current inventory, room classes, decision priorities, existing data permissions, manual survey options and source-quality records. Time basis: the proposed planning horizon and decisions that need evidence. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Owner records govern eligibility, opening, bookings, closures and approved capacity; telemetry describes anonymous aggregate observations only and cannot overwrite those facts. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, comparison and recommendation. Technique: coverage gap analysis and minimum-evidence design. Decision interval: the stated planning horizon. Minimum evidence: a complete inventory, claim-specific evidence requirements and current telemetry/authoritative-source coverage. Method: map each intended decision to required evidence, prioritise gaps by consequence and test whether authorised records or short manual counts can substitute. Uncertainty: future room changes, unknown measurement feasibility and permission constraints. Safe failure: recommend no collection where it adds little value; otherwise state the smallest verification needed before analysis; otherwise return not assessable.

ANSWER BOUNDARY

Do not assume that every room should be monitored, propose identity-based sensing, or treat lack of telemetry as low use or low importance.

WHY THIS IS USEFUL

Provides a proportionate evidence plan and makes portfolio blind spots explicit in every space-planning conclusion.

2. Occupancy, utilisation and evidence quality

Questions 9-10 of 10

Category 2 of 8

SP-019 to SP-020 | 10 questions | Use representative anonymous evidence, fair denominators and explicit quality grades rather than a single utilisation score.

SP-019

DIAGNOSTIC FOLLOW-UP, NOT CAUSE

OUTPUT: Follow-up hypotheses

EVIDENCE: Stratified evidence

SPACE-PLANNING TEAMS QUESTION

Which schedules, access conditions, room attributes or environmental conditions are associated with a persistent use pattern, and what would need checking next?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous use totals plus room-level temperature, CO2, illuminance and numerical sound indicators where proportionate and interpreted only as associations; content-free service and asset status. Spatial coverage: the named room and valid comparators, with direct room evidence only; route context is separately identified. Sampling and aggregation: 15-minute aggregates across matched periods before and during the pattern, excluding known outages and closures. Quality gate: document sensor height and representative placement, units and calibration or validation; align clocks within 60 seconds; report cadence, spatial variation, uptime, gaps, saturation, drift, maintenance and processing lineage. Readings describe sampled conditions only. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, booking/timetable, accessibility, facilities, AV/IT, cleaning, security and sensor-data owners. Records: current purpose, layout, capacity, equipment, access, opening, schedule, complaints or service tickets in aggregate, closures and known changes. Time basis: the period in which the pattern was observed and matched comparator periods. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Owner records govern eligibility, opening, bookings, closures and approved capacity; telemetry describes anonymous aggregate observations only and cannot overwrite those facts. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: comparison and diagnosis. Technique: diagnostic ranking of plausible explanations. Decision interval: the observed pattern interval. Minimum evidence: a validated pattern, current room constraints and enough comparable rooms or periods to test alternatives. Method: compare candidate factors, temporal ordering and counterexamples; rank explanations by evidential support and identify confounders without claiming causation. Uncertainty: correlated schedule and environment changes, unrecorded preferences, sparse comparators and reporting bias. Safe failure: state that no explanation is established when alternatives cannot be separated and propose targeted non-intrusive verification; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer motivation, satisfaction, health, ability, engagement or work quality, and do not attribute a pattern to a named team or cohort.

WHY THIS IS USEFUL

Directs limited investigative effort toward the most evidence-supported constraints instead of treating utilisation as self-explanatory.

SP-020

FAIR DENOMINATOR

OUTPUT: Metric specification

EVIDENCE: Owner records

SPACE-PLANNING TEAMS QUESTION

What availability window, eligible hours, exclusions and comparator make the utilisation measure fair for this decision?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional anonymous aggregate use can populate the numerator only after claim-specific quality checks. Spatial coverage: the eligible room set and periods defined by current purpose, access, services and approved operating hours. Sampling and aggregation: calculate for the selected term, year or scenario with all exclusion intervals retained. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, booking/timetable, opening-hours, accessibility, facilities, finance and sensor-data owners. Records: current room classes, capacity by layout, eligible hours, planned closures, maintenance, project possession, restricted periods and approved exceptional uses. Time basis: the reporting period and comparator cohort. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Owner records govern eligibility, opening, bookings, closures and approved capacity; telemetry describes anonymous aggregate observations only and cannot overwrite those facts. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: calculation and comparison. Technique: metric and denominator specification. Decision interval: the selected reporting period. Minimum evidence: current eligibility rules, exclusion reasons, comparable room classes and a defined numerator source. Method: calculate booked hours, occupied hours, seat-hours and intensity separately; publish denominators, exclusions, coverage and sensitivity to alternative definitions. Uncertainty: different operating models, partial-year rooms, capacity changes and missing observations. Safe failure: withhold cross-room ranking when denominators or coverage are not comparable; otherwise return not assessable.

ANSWER BOUNDARY

Do not reduce space value to one percentage, rank occupants or units, or use an arbitrary denominator to justify allocation change.

WHY THIS IS USEFUL

Creates transparent, reproducible utilisation measures that planners and owners can interpret without false precision.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

3. Capacity, layouts, equipment and suitability

Questions 1-2 of 10

Category 3 of 8

SP-021 to SP-022 | 10 questions | Match activities to approved capacity, measured layouts, installed assets and service constraints before testing change.

SP-021

APPROVED CAPACITY BY MODE

OUTPUT: Capacity lookup

EVIDENCE: Owner records

SPACE-PLANNING TEAMS QUESTION

What capacity is currently approved for this exact room, layout, activity and operating mode on the proposed date?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional furniture-position or room-setup tags that indicate which approved layout has been installed; they cannot establish capacity by themselves. Spatial coverage: the exact room and approved layout variant, including separately controlled sub-areas. Sampling and aggregation: event-driven setup state for the requested interval, with a current verification timestamp. Quality gate: verify asset identity, room or service mapping and state semantics through commissioning or stocktake; retain event time, stale-state timeout, expected heartbeat, uptime, gaps, relocations, duplicates, firmware and source lineage. A heartbeat is not proof of suitability. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: capacity, fire-safety, accessibility, room, facilities and activity owners. Records: current approved capacity schedule, layout drawings, furniture plan, egress and accessible-clearance constraints, temporary restrictions and permitted operating modes. Time basis: the requested date and layout, using approvals effective then. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current capacity, layout, asset and competent technical records govern; telemetry may corroborate state but cannot certify suitability, safety or accessibility. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup. Technique: capacity-by-layout lookup. Decision interval: the proposed use interval. Minimum evidence: a verified room, proposed layout and current approved capacity record for that layout. Method: match the proposed or installed layout to the authoritative schedule and list any mode-specific conditions. Uncertainty: unrecorded furniture moves, temporary restrictions and layouts that are similar but not identical. Safe failure: state capacity unknown and require owner confirmation when the exact layout cannot be matched; otherwise return not assessable.

ANSWER BOUNDARY

Do not calculate an approved capacity from floor area alone, certify safe occupancy, or treat observed headcount as permission to exceed the current limit.

WHY THIS IS USEFUL

Ensures that every demand or scenario calculation uses the capacity applicable to the actual authorised layout.

SP-022

ACTIVITY-SPACE FIT

OUTPUT: Eligible-room list

EVIDENCE: Requirements + records

SPACE-PLANNING TEAMS QUESTION

Which current spaces satisfy every declared hard requirement for this activity without works, fixed-asset relocation or policy exception?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: content-free status for required AV, network, power, ventilation, access and specialist assets; optional room-level environmental summaries where the activity defines numerical limits. Spatial coverage: candidate rooms and their directly serving assets only; shared services and upstream dependencies remain explicit. Sampling and aggregation: current status plus the full proposed activity interval, using event times and health heartbeats. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: activity owner, space and room owners, booking/timetable, capacity, accessibility, facilities, IT/AV, safety and specialist-service owners. Records: activity requirements, eligible room types, capacity/layout, equipment, access, service levels, restrictions, opening, bookings and closures. Time basis: the proposed activity interval and setup/recovery windows. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current capacity, layout, asset and competent technical records govern; telemetry may corroborate state but cannot certify suitability, safety or accessibility. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, comparison and recommendation. Technique: hard-constraint filtering and suitability comparison. Decision interval: the proposed activity plus setup and recovery. Minimum evidence: an owner-approved requirement set and current authoritative constraints for every candidate room. Method: exclude any candidate failing a hard constraint; compare remaining spaces on declared preferences and evidence confidence without inventing missing capability. Uncertainty: uncertain equipment health, unresolved accessibility evidence, shared-service limits and future changes. Safe failure: return no eligible option when any critical requirement lacks current evidence; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer activity requirements from content, promise that equipment will work, certify accessibility or safety, or book or assign a room.

WHY THIS IS USEFUL

Gives planners a valid candidate set before they consider cost, proximity or utilisation trade-offs.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

3. Capacity, layouts, equipment and suitability

SP-023 to SP-024 | 10 questions | Match activities to approved capacity, measured layouts, installed assets and service constraints before testing change.

Questions 3-4 of 10

Category 3 of 8

SP-023

INSTALLED STATE VALIDATION

OUTPUT: Discrepancy list

EVIDENCE: Asset status + survey

SPACE-PLANNING TEAMS QUESTION

Where do the installed furniture, equipment, service points or room setup differ from current approved layout and asset records?

SENSORS AND TELEMETRY

Proposed modality and location: proposed furniture or asset tags, non-content device heartbeats, power-state metadata and authorised manual setup confirmation; no cameras or screen inspection. Spatial coverage: the named room, each tagged movable item and fixed asset included in the approved layout. Sampling and aggregation: event-driven changes with a pre-use snapshot; aggregate only item-presence status, not user interaction. Quality gate: verify asset identity, room or service mapping and state semantics through commissioning or stocktake; retain event time, stale-state timeout, expected heartbeat, uptime, gaps, relocations, duplicates, firmware and source lineage. A heartbeat is not proof of suitability. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: room and space owners, furniture/asset register owner, AV/IT, facilities, accessibility and capacity owners. Records: current approved layout, furniture and equipment inventory, asset locations, setup instructions, temporary loans, maintenance and approved changes. Time basis: the intended use interval and latest completed setup event. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current capacity, layout, asset and competent technical records govern; telemetry may corroborate state but cannot certify suitability, safety or accessibility. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation and comparison. Technique: installed-state versus approved-state comparison. Decision interval: the pre-use or planning snapshot. Minimum evidence: the approved room configuration and current item-level or manual setup evidence. Method: compare required, present, missing, extra and uncertain items; distinguish location mismatch from service health and preserve all exclusions. Uncertainty: untagged furniture, moved assets, dead tags, temporary loans and incomplete manual confirmation. Safe failure: classify the setup as unverified rather than complete when any critical item is unknown; otherwise return not assessable.

ANSWER BOUNDARY

Do not inspect device content, operate equipment, move furniture, certify the layout or infer who changed it.

WHY THIS IS USEFUL

Highlights configuration evidence that must be corrected before capacity or suitability is relied upon.

SP-024

LAYOUT OPTION COMPARISON

OUTPUT: Layout comparison

EVIDENCE: Geometry + scenario

SPACE-PLANNING TEAMS QUESTION

For each feasible layout, how do usable capacity, circulation, clearances, sightlines, teaching or work positions and service reach change?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional current measured geometry from authorised laser or manual survey, furniture dimensions and non-personal door or service-location data. Spatial coverage: the exact room boundary, fixed obstructions, doors, circulation paths, furniture and service points. Sampling and aggregation: a dated survey and scenario version; no continuous monitoring is needed. Quality gate: retain survey method, instrument calibration or validation, survey date, coordinate reference, tolerance, full boundary or route coverage, obscured areas, revision and competent provenance. A survey is a dated observation, not approval. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: architectural/space information owner, capacity, fire-safety, accessibility, facilities, room and specialist- service owners. Records: current measured survey, approved layouts, egress and clearance criteria, furniture/equipment dimensions, service locations and proposal constraints. Time basis: the proposed change date and scenario version. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current capacity, layout, asset and competent technical records govern; telemetry may corroborate state but cannot certify suitability, safety or accessibility. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: calculation and simulation. Technique: geometry-based scenario simulation. Decision interval: the stated layout scenario. Minimum evidence: current measured geometry, exact furniture/equipment model and authoritative constraints. Method: place scenario elements, calculate clearances, routes and seat positions, identify constraint breaches and show sensitivity to dimensional tolerance. Uncertainty: survey age, movable obstructions, user-specific needs and approval criteria that require competent interpretation. Safe failure: label the scenario indicative and withhold feasibility when a critical dimension or rule is missing; otherwise return not assessable.

ANSWER BOUNDARY

Do not approve the layout, certify capacity, accessibility, fire safety or ergonomics, or instruct physical changes.

WHY THIS IS USEFUL

Allows planners to discard obviously infeasible options before commissioning detailed professional design.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

3. Capacity, layouts, equipment and suitability

Questions 5-6 of 10

Category 3 of 8

SP-025 to SP-026 | 10 questions | Match activities to approved capacity, measured layouts, installed assets and service constraints before testing change.

SP-025

RECONFIGURATION BLOCKERS

OUTPUT: Constraint register

EVIDENCE: Survey + owner records

SPACE-PLANNING TEAMS QUESTION

Which fixed elements, structural limits, service zones, compliance dependencies and operating rules constrain reconfiguration of this room?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional position and state metadata for fixed partitions, doors, floor boxes, radiators, sensors, equipment and service outlets. Spatial coverage: the room boundary and directly relevant fixed or semi-fixed elements; concealed services require authoritative drawings or survey. Sampling and aggregation: current snapshot plus effective dates for authorised works. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: building information, facilities, M&E, IT/AV, accessibility, fire-safety, room and asset owners. Records: current measured survey, structural and services information, fixed furniture, outlets, doors, partitions, ceiling/floor constraints, warranties and approved works. Time basis: the proposed intervention horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current capacity, layout, asset and competent technical records govern; telemetry may corroborate state but cannot certify suitability, safety or accessibility. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: constraint inventory and feasibility screening. Decision interval: the stated reconfiguration scenario. Minimum evidence: verified current geometry and owner-confirmed fixed/service constraints. Method: classify elements as fixed, relocatable with approval, capacity-limited or unknown; identify which scenario assumptions touch each constraint. Uncertainty: concealed conditions, outdated surveys, undocumented local changes and service capacities not tested under the scenario. Safe failure: exclude uncertain zones or require a competent survey before feasibility is claimed; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer structure or concealed services from surface sensing, authorise intrusive work, or promise that a relocatable item can be moved.

WHY THIS IS USEFUL

Shows where a seemingly simple layout change actually depends on technical surveys, service capacity or owner consent.

SP-026

DIGITAL AND HYBRID READINESS

OUTPUT: Readiness matrix

EVIDENCE: Service telemetry + records

SPACE-PLANNING TEAMS QUESTION

Which rooms have verified current AV, network, power, lighting and acoustic support for the stated digital or hybrid activity, and where is reliability risk material?

SENSORS AND TELEMETRY

Proposed modality and location: content-free AV device health, network service metrics, outlet/power status and numerical background sound or lighting indicators; no packet, screen, media or speech inspection. Spatial coverage: the named room and directly serving network, AV and power assets, with shared upstream dependencies explicit. Sampling and aggregation: health events plus five- minute service summaries over comparable activity periods and the proposed window. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: activity owner, room/space, AV, IT/network, facilities, accessibility and booking owners. Records: current activity requirements, installed equipment, supported configurations, network/service commitments, capacity/layout, opening, bookings and known incidents. Time basis: the proposed activity and setup interval. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current capacity, layout, asset and competent technical records govern; telemetry may corroborate state but cannot certify suitability, safety or accessibility. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, observation and comparison. Technique: hard-constraint screening plus reliability comparison. Decision interval: the proposed activity interval. Minimum evidence: owner-approved technical requirements and current room/service evidence for each candidate. Method: exclude unsupported configurations, compare eligible rooms using historical service reliability and environmental context, and report fallback dependencies. Uncertainty: future incidents, shared-network load, device configuration changes and subjective acoustic experience. Safe failure: state capability unconfirmed when any critical service lacks current owner evidence or valid health data; otherwise return not assessable.

ANSWER BOUNDARY

Do not inspect content or credentials, guarantee connectivity or AV performance, change configuration, or certify the room for a specific person's needs.

WHY THIS IS USEFUL

Supports credible space selection and identifies where a service test or local support arrangement is necessary.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

3. Capacity, layouts, equipment and suitability

Questions 7-8 of 10
Category 3 of 8

SP-027 to SP-028 | 10 questions | Match activities to approved capacity, measured layouts, installed assets and service constraints before testing change.

SP-027

SPECIALIST EQUIPMENT RELOCATION

OUTPUT: Move-feasibility gate

EVIDENCE: Surveys + asset records

SPACE-PLANNING TEAMS QUESTION

Can the specified equipment be moved to, installed in and operated within a candidate room using verified routes, dimensions, loads, utilities and downtime windows?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: owner-issued lift/door state, route closure status and content-free service-point availability; optional equipment location tag only where authorised. Spatial coverage: the approved origin, candidate destination, full movement route and directly required services. Sampling and aggregation: current status for the planned move window, with event-driven changes and a recent route verification. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: equipment owner, logistics/moves, space, facilities, lift, security/access, M&E; safety and route/accessibility owners. Records: equipment dimensions, mass and handling constraints, route and door/lift dimensions, load limits, access conditions, permits, service requirements and planned closures. Time basis: the proposed move and operating horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current capacity, layout, asset and competent technical records govern; telemetry may corroborate state but cannot certify suitability, safety or accessibility. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, calculation and comparison. Technique: constrained route and service feasibility calculation. Decision interval: the planned move window and destination use period. Minimum evidence: verified equipment envelope, current route geometry/status and destination service capacity. Method: test every route segment and destination constraint with tolerances; identify the first unresolved or failed constraint and alternative eligible routes. Uncertainty: temporary obstructions, equipment packaging, lift availability and service capacity not measured under load. Safe failure: return route or operational feasibility as unconfirmed and defer to competent logistics/technical owners; otherwise return not assessable.

ANSWER BOUNDARY

Do not direct lifting or handling, authorise access or permits, operate doors or lifts, or declare equipment safe to move or use.

WHY THIS IS USEFUL

Prevents candidate spaces from being selected when equipment cannot be delivered, installed or supported.

SP-028

STALE SUITABILITY EVIDENCE

OUTPUT: Survey priority

EVIDENCE: Record assurance

SPACE-PLANNING TEAMS QUESTION

Which capacity, layout, equipment or suitability records are too old, incomplete, spatially invalid or conflicting for the decision, and what should be surveyed first?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: asset and room metadata showing last commissioning, inspection, relocation, maintenance and source-health dates. Spatial coverage: every room, layout, asset and service required by the proposed decision. Sampling and aggregation: assessed at the decision date, with age compared against owner-defined validity rules. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: capacity, accessibility, fire-safety, room, asset, AV/IT, facilities and building-information owners. Records: approved capacities/layouts, measured surveys, equipment inventories, service tests, restrictions, planned works and validity or revalidation rules. Time basis: the requested present or future decision interval. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current capacity, layout, asset and competent technical records govern; telemetry may corroborate state but cannot certify suitability, safety or accessibility. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: record-validity and evidence-gap classification. Decision interval: the decision interval. Minimum evidence: the complete set of claims required for the decision plus owner-defined validity criteria. Method: compare each record's scope, effective date, observation date and known changes with the decision; classify current, conditionally usable, conflicting or insufficient. Uncertainty: unknown physical changes, records with broad scope and validity rules that differ by domain. Safe failure: remove unsupported claims and state the exact survey, owner confirmation or service test needed; otherwise return not assessable.

ANSWER BOUNDARY

Do not extend a record beyond its scope or validity, infer capacity from a previous layout, or present missing evidence as a minor administrative issue.

WHY THIS IS USEFUL

Focuses limited survey and verification effort on the evidence that actually blocks a space decision.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

3. Capacity, layouts, equipment and suitability

Questions 9-10 of 10

Category 3 of 8

SP-029 to SP-030 | 10 questions | Match activities to approved capacity, measured layouts, installed assets and service constraints before testing change.

SP-029

STANDARD ROOM FAMILY

OUTPUT: Standardisation candidates

EVIDENCE: Asset + requirement comparison

SPACE-PLANNING TEAMS QUESTION

Which comparable rooms could use a standard furniture, equipment or service kit without removing specialist, inclusive or locally essential capability?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: current furniture/equipment inventories, measured dimensions and optional asset-position tags; no occupant sensing is required. Spatial coverage: candidate rooms with comparable current purpose and exact geometry/service constraints. Sampling and aggregation: a dated inventory and survey snapshot for the planning horizon. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, room, furniture, asset, accessibility, capacity, fire-safety, AV/IT and facilities owners. Records: current room types, approved layouts, activity requirements, furniture/equipment models, service points, inclusive requirements and planned changes. Time basis: the proposed standardisation horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current capacity, layout, asset and competent technical records govern; telemetry may corroborate state but cannot certify suitability, safety or accessibility. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: comparison, simulation and recommendation. Technique: constraint-aware clustering and scenario comparison. Decision interval: the proposed standardisation horizon. Minimum evidence: current geometry, approved constraints and owner-defined minimum functionality for all candidate rooms. Method: cluster only rooms that share hard constraints, test a standard kit in each room and report exceptions, lost capabilities and reversible variants. Uncertainty: hidden local requirements, ageing inventories and future changes that break comparability. Safe failure: exclude any room whose critical requirement or current geometry is uncertain; otherwise return not assessable.

ANSWER BOUNDARY

Do not erase specialist or inclusive needs to improve apparent consistency, purchase or move assets, or approve a standard layout.

WHY THIS IS USEFUL

Identifies where standard kits may simplify moves and maintenance while preserving room-specific constraints.

SP-030

CHANGE-OF-USE DUE DILIGENCE

OUTPUT: Due-diligence pack

EVIDENCE: Records + consultation

SPACE-PLANNING TEAMS QUESTION

What evidence, consultation, surveys and competent approvals must be assembled before this change-of-use option can enter formal consideration?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: only proportionate contextual measurements relevant to the proposed use, such as room-level environmental baselines or service capacity; sensing cannot establish legal or technical approval. Spatial coverage: the exact room, proposed use, affected routes, shared facilities and service zones. Sampling and aggregation: baseline periods and scenario dates defined by the competent owners. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: institutional space authority, planning/building control as applicable, fire-safety, accessibility, M&E, facilities, security, room, academic/activity and information owners. Records: current purpose and title, measured survey, capacity/layout, access/egress, services, hazards, restrictions, consents, demand case, consultation inputs and project constraints. Time basis: the proposed change horizon and applicable rule set. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current capacity, layout, asset and competent technical records govern; telemetry may corroborate state but cannot certify suitability, safety or accessibility. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and recommendation. Technique: evidence-requirement classification and gap analysis. Decision interval: the pre-proposal stage. Minimum evidence: a defined proposed use, exact spatial boundary and the owner-issued requirement checklist. Method: map every proposed-use requirement to current evidence and accountable owner; classify satisfied, conditional, missing or outside the service's competence. Uncertainty: requirements that depend on professional judgement, user consultation, intrusive survey or future design. Safe failure: provide a bounded evidence checklist and stop before feasibility or approval where critical items remain unresolved; otherwise return not assessable.

ANSWER BOUNDARY

Do not certify change-of-use feasibility, interpret law or technical standards beyond owner records, authorise design or works, or replace consultation and measured surveys.

WHY THIS IS USEFUL

Gives planners a transparent route from early idea to the evidence and professional inputs needed for an accountable proposal.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

4. Adjacencies, circulation and shared facilities

SP-031 to SP-032 | 10 questions | Plan relationships, journeys and shared facilities as a connected network, not as isolated rooms.

Questions 1-2 of 10

Category 4 of 8

SP-031

CRITICAL ADJACENCIES

OUTPUT: Adjacency requirements

EVIDENCE: Consultation + records

SPACE-PLANNING TEAMS QUESTION

Which functions must be co-located or kept within a stated travel time, which are preferences, and which adjacency claims lack evidence?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional anonymous zone-level flow totals between broad areas, without device linkage or individual journey histories. Spatial coverage: only the named functions, candidate areas and authorised connecting routes; no person-level origins or destinations. Sampling and aggregation: 15-minute area totals across representative operating weeks where proportionate. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space and activity owners, local service leads, accessibility, facilities, security and building-information owners. Records: authorised functional requirements, shared-resource dependencies, service workflows, confidentiality/security constraints, current locations and consultation evidence. Time basis: the planning horizon and operating model being considered. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current route, service, safety, booking and facility-owner records govern; observed flows may reveal pressure but cannot authorise routing or facility change. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: constraint classification and adjacency-graph analysis. Decision interval: the stated scenario horizon. Minimum evidence: owner-confirmed functions, dependencies and a current route graph. Method: classify each adjacency as hard, service-critical, beneficial or unverified; model distance and shared dependencies without treating co-movement as need. Uncertainty: informal working practices, future operating changes and stated preferences that vary between users. Safe failure: retain unresolved adjacencies as consultation items and do not optimise them as facts; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer collaboration, team relationships or activity importance from proximity or flow, disclose sensitive functions, or decide which groups should be colocated.

WHY THIS IS USEFUL

Creates a verified adjacency brief so scenarios preserve mandatory service relationships without treating preferences as facts.

SP-032

SHARED-FACILITY CATCHMENT

OUTPUT: Catchment pressure map

EVIDENCE: Records + aggregate demand

SPACE-PLANNING TEAMS QUESTION

Which rooms depend on shared toilets, kitchens, lockers, print, storage, meeting, technical-support or welfare facilities, and where could demand exceed provision?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous area counts, doorway totals, fill-level or service-status telemetry where commissioned, and non-content equipment heartbeats. Spatial coverage: each named shared facility, its access route and the rooms it is authorised to serve; no individual linkage between origin and destination. Sampling and aggregation: one-minute counts aggregated to 15-minute demand bands, with event logs for outages and replenishment where applicable. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, facilities, cleaning, waste, IT/print, accessibility, security and local service owners. Records: current shared-facility inventory, service areas, capacities, opening/access rules, maintenance status, replenishment rounds, closures and planned changes. Time basis: the selected operating period and scenario. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current route, service, safety, booking and facility-owner records govern; observed flows may reveal pressure but cannot authorise routing or facility change. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, observation and comparison. Technique: dependency mapping and demand-band comparison. Decision interval: the selected operating period. Minimum evidence: current service boundaries, eligible users at aggregate area level and valid facility demand evidence. Method: build a room-to-facility dependency graph, compare demand distributions with owner-defined service assumptions and identify single points of pressure. Uncertainty: unrecorded informal facilities, shared routes, sensor saturation and demand displaced from closed amenities. Safe failure: show only covered dependencies and defer service adequacy to the responsible owner; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer personal use of toilets, kitchens or welfare facilities, certify hygiene or accessibility, or direct service staffing or replenishment.

WHY THIS IS USEFUL

Shows whether a space scenario transfers hidden demand onto already constrained shared facilities.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

4. Adjacencies, circulation and shared facilities

Questions 3-4 of 10

Category 4 of 8

SP-033 to SP-034 | 10 questions | Plan relationships, journeys and shared facilities as a connected network, not as isolated rooms.

SP-033

CIRCULATION PINCH POINTS

OUTPUT: Pinch-point schedule

EVIDENCE: Count-only route evidence

SPACE-PLANNING TEAMS QUESTION

Which corridor, doorway, stair or lift-lobby segments repeatedly constrain normal transitions, and during which planning-relevant periods?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous bidirectional counters, ceiling-zone density totals and door/lift state on covered public or authorised routes; no cameras, faces or journey linkage. Spatial coverage: each named corridor segment, doorway, stair or lift lobby and its adjacent decision points; uncovered branches remain explicit. Sampling and aggregation: one-minute aggregates around matched transitions across at least four representative weeks. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, timetable/event, circulation, accessibility, facilities, lift/door, security and fire-safety owners. Records: current route graph, room transition times, door/lift status, closures, route restrictions, approved events and any temporary diversions. Time basis: the selected transitions and current route configuration. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current route, service, safety, booking and facility-owner records govern; observed flows may reveal pressure but cannot authorise routing or facility change. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation and comparison. Technique: spatial-temporal hotspot analysis. Decision interval: the selected transition windows. Minimum evidence: valid aggregate coverage of the full candidate bottleneck and current schedule/route context. Method: calculate duration and recurrence above agreed planning bands, compare matched transitions and test whether alternative routes explain redistribution. Uncertainty: counter overlap, waiting versus through-flow ambiguity, temporary obstructions and incomplete route coverage. Safe failure: describe only the observed covered segment and refer safety significance to competent owners; otherwise return not assessable.

ANSWER BOUNDARY

Do not track people across route segments, infer cohort behaviour, certify evacuation or crowd safety, or control doors, lifts, signage or movement.

WHY THIS IS USEFUL

Supports targeted layout, scheduling or route studies instead of relying on anecdotal congestion claims.

SP-034

QUEUE SPILLOVER

OUTPUT: Spillover action list

EVIDENCE: Counts + schedule records

SPACE-PLANNING TEAMS QUESTION

Where do queues or waiting areas repeatedly encroach on circulation, neighbouring activity or accessible routes, and what generates the peak window?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous zone occupancy totals, doorway counts and numerical sound levels without recording speech. Spatial coverage: the named service point, approved waiting area and adjacent route segments; no identity or path continuity. Sampling and aggregation: one-minute aggregates during service/event windows, summarised by comparable day and demand context. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, event, reception/service, circulation, accessibility, security and facilities owners. Records: current waiting-area designation, service hours, bookings/events, route width/status, queue management arrangements, closures and temporary setups. Time basis: the selected service or event periods. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current route, service, safety, booking and facility-owner records govern; observed flows may reveal pressure but cannot authorise routing or facility change. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation and comparison. Technique: zone occupancy and recurrence comparison. Decision interval: the selected service or event windows. Minimum evidence: valid aggregate coverage of waiting and route zones plus authoritative service/event timing. Method: compare occupied-area bands and duration with owner-defined planning thresholds; separate scheduled peaks, exceptional events and unknown causes. Uncertainty: sensor inability to classify waiting, mixed through-flow and temporary furniture changes. Safe failure: state that spillover is not established when zone boundaries or context are insufficient; otherwise return not assessable.

ANSWER BOUNDARY

Do not identify people in a queue, infer the service they seek or their waiting experience, certify route safety, or direct crowd operations.

WHY THIS IS USEFUL

Helps planners test whether waiting areas, entrances or service locations need spatial change.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

4. Adjacencies, circulation and shared facilities

Questions 5-6 of 10

Category 4 of 8

SP-035 to SP-036 | 10 questions | Plan relationships, journeys and shared facilities as a connected network, not as isolated rooms.

SP-035

INCOMPATIBLE ADJACENCIES

OUTPUT: Conflict matrix

EVIDENCE: Requirements + measurements

SPACE-PLANNING TEAMS QUESTION

Which existing or proposed adjacencies create credible conflicts in noise, vibration, privacy, security, contamination control, deliveries or building services?

SENSORS AND TELEMETRY

Proposed modality and location: numerical sound statistics, vibration, temperature/air indicators and content-free door or service state where proportionate; no audio, video or content capture. Spatial coverage: the source room, receiving room, shared boundary and relevant service path; corridor data cannot replace receiving-room evidence. Sampling and aggregation: one- or five-minute observations aggregated by matched activity periods, with event timestamps for doors and services. Quality gate: document sensor height and representative placement, units and calibration or validation; align clocks within 60 seconds; report cadence, spatial variation, uptime, gaps, saturation, drift, maintenance and processing lineage. Readings describe sampled conditions only. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space and activity owners, acoustic/environmental specialists, security, safety, facilities, laboratory or specialist-service and accessibility owners. Records: current room purposes, confidentiality/security classifications, hazard/service requirements, construction details, complaints in aggregate, schedules and approved limits. Time basis: the current pattern or stated scenario interval. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current route, service, safety, booking and facility-owner records govern; observed flows may reveal pressure but cannot authorise routing or facility change. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: comparison and diagnosis. Technique: constraint conflict analysis and diagnostic comparison. Decision interval: the current or proposed adjacency interval. Minimum evidence: owner-confirmed requirements, current boundary/service evidence and comparable control periods. Method: test each claimed conflict against direct evidence, temporal alignment and plausible transmission paths; distinguish confirmed constraint, association and unknown. Uncertainty: subjective response, multiple sources, intermittent activities and incomplete construction information. Safe failure: avoid causal or compliance conclusions and specify the competent survey or consultation needed; otherwise return not assessable.

ANSWER BOUNDARY

Do not inspect confidential activities, infer who caused a disturbance, disclose restricted functions, or certify acoustic, privacy, security or contamination control.

WHY THIS IS USEFUL

Prevents a relocation from solving one demand problem while creating an incompatible neighbour relationship.

SP-036

CLOSURE REDISTRIBUTION

OUTPUT: Redistribution estimate

EVIDENCE: Records + aggregate demand

SPACE-PLANNING TEAMS QUESTION

If a shared room or facility closes, where is displaced demand likely to move and which nearby alternatives could become overloaded?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous room/zone counts and route totals with no cross-sensor identifiers; owner-issued closure and service-status events. Spatial coverage: the closed facility, authorised alternatives and connecting route zones; no attempt to follow individuals. Sampling and aggregation: one-minute observations aggregated to 15-minute bands before, during and after comparable closures. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, room/facility, booking/timetable, event, facilities, accessibility and security owners. Records: closure scope and times, authorised alternatives, current room purpose/capacity, bookings, events, routes and other simultaneous disruptions. Time basis: the closure interval and matched comparator periods. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current route, service, safety, booking and facility-owner records govern; observed flows may reveal pressure but cannot authorise routing or facility change. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation, comparison and estimate. Technique: before-during comparison and redistribution estimate. Decision interval: the closure interval. Minimum evidence: authoritative closure record plus valid aggregate coverage of plausible alternatives and matched baseline periods. Method: compare area-level demand changes, test whether totals balance within uncertainty and distinguish observed redistribution from unobserved or reduced demand. Uncertainty: simultaneous schedule changes, incomplete alternative coverage and people leaving the building. Safe failure: report only aggregate associations and state when the destination of displaced demand is unknown; otherwise return not assessable.

ANSWER BOUNDARY

Do not reconstruct individual journeys, infer why people moved, or recommend use of an alternative that is not currently eligible and owner-confirmed.

WHY THIS IS USEFUL

Shows the wider portfolio impact of losing a shared facility and helps test contingency assumptions.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

4. Adjacencies, circulation and shared facilities

SP-037 to SP-038 | 10 questions | Plan relationships, journeys and shared facilities as a connected network, not as isolated rooms.

Questions 7-8 of 10

Category 4 of 8

SP-037PUBLIC AND SERVICE-ROUTE SEPARATIONOUTPUT: Route-separation optionsEVIDENCE: Schedule + route evidence

SPACE-PLANNING TEAMS QUESTIONWhere do public journeys conflict with delivery, maintenance, waste, equipment-move or event-setup routes, and can timing or routing separate them?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous route counts, door/lift state and owner-issued logistics status; no worker or visitor identification. Spatial coverage: the approved public and service routes, shared segments, staging areas and access points. Sampling and aggregation: one-minute aggregates during authorised logistics windows and matched public-use periods; event- driven closure/status. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, logistics, facilities, events, security/access, accessibility, health-and-safety and route owners. Records: current delivery/maintenance/setup schedules, approved routes and staging, public opening/events, restrictions, permits, closures and alternative paths. Time basis: the candidate logistics and public-use intervals. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current route, service, safety, booking and facility-owner records govern; observed flows may reveal pressure but cannot authorise routing or facility change. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, comparison and simulation. Technique: schedule overlap and route-conflict comparison. Decision interval: the candidate operating windows. Minimum evidence: current authorised routes and schedules plus valid aggregate demand on shared segments. Method: identify time-space overlaps, compare eligible windows or routes and expose unresolved access, accessibility or safety dependencies. Uncertainty: late logistics changes, temporary obstructions and incomplete route observations. Safe failure: provide only eligible comparison options and defer operational control to the responsible authorities; otherwise return not assessable.

ANSWER BOUNDARY

Do not disclose sensitive logistics detail beyond permission, direct deliveries, waive permits, certify route safety, or control access and barriers.

WHY THIS IS USEFULAllows planners to identify where a space or event scenario creates avoidable conflict with essential building operations.

SP-038SINGLE POINTS OF FAILUREOUTPUT: Dependency risk mapEVIDENCE: Owner records + status

SPACE-PLANNING TEAMS QUESTIONWhich spaces depend on one lift, door, route, utility or shared facility, and what functions lose a usable alternative if it fails?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: owner-issued lift, door and shared-service state with health heartbeat; optional non-personal route status. Spatial coverage: every candidate space, all authorised routes and the directly required shared services. Sampling and aggregation: current status and scenario outage intervals, preserving effective and event times. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, accessibility, lift/door, facilities, security/access, fire-safety and shared-service owners. Records: current route graph, access rules, lift/door/service dependencies, alternative routes, closures, capacity and contingency arrangements. Time basis: the stated outage and planning horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current route, service, safety, booking and facility-owner records govern; observed flows may reveal pressure but cannot authorise routing or facility change. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and simulation. Technique: dependency-graph and single-failure analysis. Decision interval: the stated outage scenario. Minimum evidence: current topology and authoritative eligibility of every alternative path/service. Method: remove each dependency in turn, calculate spaces or functions disconnected or degraded and show whether alternatives satisfy all hard constraints. Uncertainty: status changes, hidden dependencies and user-specific requirements not represented in the graph. Safe failure: mark impacts unknown where an alternative's current accessibility or service status is unverified; otherwise return not assessable.

ANSWER BOUNDARY

Do not declare an actual closure, certify emergency resilience or accessibility, or instruct use of routes not authorised for the purpose.

WHY THIS IS USEFULReveals fragile clusters that may need alternative access, service redundancy or contingency space.

4. Adjacencies, circulation and shared facilities

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Category 4 of 8

SP-039 to SP-040 | 10 questions | Plan relationships, journeys and shared facilities as a connected network, not as isolated rooms.

SP-039

SHARED-FACILITY LOCATION

OUTPUT: Eligible-location set

EVIDENCE: Constraints + scenario

SPACE-PLANNING TEAMS QUESTION

Which eligible locations satisfy the mandatory catchment, route, inclusion, capacity, service and resilience constraints for a proposed shared facility?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous area-level demand totals and route counts; optional facility-specific service capacity or environmental indicators. Spatial coverage: candidate locations, authorised origin zones and all eligible routes, using aggregates large enough to protect privacy. Sampling and aggregation: 15-minute demand bands across representative periods and bounded future scenarios. Quality gate: document sensor height and representative placement, units and calibration or validation; align clocks within 60 seconds; report cadence, spatial variation, uptime, gaps, saturation, drift, maintenance and processing lineage. Readings describe sampled conditions only. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, facility/service, accessibility, security, facilities, activity and building-information owners. Records: candidate-space eligibility, route distances and constraints, service requirements, current shared- facility distribution, aggregate demand assumptions and planned changes. Time basis: the approved planning horizon and demand scenarios. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current route, service, safety, booking and facility-owner records govern; observed flows may reveal pressure but cannot authorise routing or facility change. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: simulation, optimisation and recommendation. Technique: multi-criteria location optimisation. Decision interval: the stated future horizon. Minimum evidence: eligible candidate spaces, owner-approved demand zones, current routes and hard service/access constraints. Method: minimise aggregate travel and route pressure while preserving inclusive alternatives, service feasibility and resilience; show trade-offs and sensitivity. Uncertainty: future demand distribution, unrepresented user needs and incomplete route evidence. Safe failure: return a non-ranked feasible set when preferences or evidence are insufficient for a defensible ordering; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer individual origins, choose a location on behalf of owners, certify inclusive access, or displace existing uses automatically.

WHY THIS IS USEFUL

Supports transparent location choices for shared amenities, storage, support or collaboration facilities.

SP-040

ADJACENCY AND CIRCULATION TRADE-OFFS

OUTPUT: Scenario comparison

EVIDENCE: Governed scenario

SPACE-PLANNING TEAMS QUESTION

Across feasible regrouping options, how do adjacency quality, journey length, route pressure, shared-facility demand, noise exposure and dependencies change?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous area-level use and route totals, numerical environmental indicators and content-free service status where proportionate. Spatial coverage: current and candidate zones, eligible routes, shared facilities and directly affected boundaries; no individual journey reconstruction. Sampling and aggregation: 15-minute aggregates across representative baselines, projected to stated scenario intervals. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space and activity owners, building information, accessibility, facilities, security, timetable/event and specialist-service owners. Records: current locations and demand, proposed placements, route graph, shared-service capacities, adjacency constraints, planned changes and consultation inputs. Time basis: the scenario horizon and operating assumptions. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current route, service, safety, booking and facility-owner records govern; observed flows may reveal pressure but cannot authorise routing or facility change. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: simulation and comparison. Technique: spatial demand redistribution simulation. Decision interval: the stated scenario horizon. Minimum evidence: a complete scenario, current baseline, valid routes and owner-approved demand/constraint assumptions. Method: redistribute aggregate demand, calculate changes in route load, distance, shared service pressure and environmental exposure, and compare multiple feasible options. Uncertainty: behavioural adaptation, uncertain future demand, unmodelled preferences and causal limits. Safe failure: show ranges and unsupported criteria; do not produce a single preferred scenario when evidence or consultation is incomplete; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer collaboration or satisfaction, promise improved outcomes, decide relocations, or treat modelled flows as observed future behaviour.

WHY THIS IS USEFUL

Makes the consequences of regrouping visible across the whole building rather than only at destination rooms.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

Stakeholder Question Catalogue 29 | Category 4: Adjacencies, circulation and shared facilities

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5. Accessibility, inclusion and diverse user needs

SP-041 to SP-042 | 10 questions | Preserve choice, dignity and workable alternatives through consultation, current surveys and competent assessment.

Questions 1-2 of 10

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SP-041

ACCESSIBLE-JOURNEY EVIDENCE

OUTPUT: Evidence-status map

EVIDENCE: Access audit + status

SPACE-PLANNING TEAMS QUESTION

What current authoritative evidence describes the complete accessible journey to, into and within each candidate space during the relevant operating period?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: owner-issued lift, automatic- door and route-status heartbeats where commissioned; sensing cannot establish accessibility. Spatial coverage: entrance, external approach where owned, doors, lifts or stairs, circulation, destination, seating and relevant amenities. Sampling and aggregation: current status for the requested interval, with event times and a recent owner update. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: accessibility/inclusion, Estates access, lift/door, space, facilities, security and emergency-information owners. Records: current access descriptions, measured route/door/lift information, designated facilities, seating and assistive provision, opening, closures, restrictions and planned changes. Time basis: the requested present or future interval. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current access audits, user-stated requirements, service status and competent accessibility or fire determinations govern; sensing cannot certify inclusion. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup. Technique: whole-journey evidence lookup and completeness classification. Decision interval: the requested access interval. Minimum evidence: a verified origin/entrance, destination and current owner evidence for every route segment and relevant facility. Method: assemble the journey, classify each segment as confirmed, status-dependent, missing or conflicting and preserve alternative routes. Uncertainty: temporary obstructions, incomplete measurements, changing door/lift status and needs not represented in records. Safe failure: state exactly which segment is unconfirmed and refer to the accessibility owner rather than claiming whole-route suitability; otherwise return not assessable.

ANSWER BOUNDARY

Do not diagnose a person, infer disability, certify accessibility, promise independent access, or present a route as suitable without complete current evidence.

WHY THIS IS USEFUL

Shows where space options are genuinely assessable and where missing route evidence requires owner action or user consultation.

SP-042

FUNCTIONAL REQUIREMENT MATCH

OUTPUT: Suitable-option set

EVIDENCE: Consented requirements

SPACE-PLANNING TEAMS QUESTION

Which candidate spaces meet every voluntarily stated functional requirement for the activity, time, route and support arrangement?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: only status or numerical evidence relevant to requirements the person chooses to state, such as lift/door state, light, sound, temperature or CO2. Spatial coverage: the complete eligible journey and named candidate spaces; no collection beyond the current decision. Sampling and aggregation: current or forecast interval with one- to 15-minute aggregates as appropriate and a stated validity window. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: the person for their chosen functional requirements; accessibility, space, room, facilities, lift/door, booking and activity owners for current provision. Records: current route/access features, room layout, seating, assistive provision, amenities, opening, bookings, closures and support arrangements. Time basis: the stated activity and travel interval. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current access audits, user-stated requirements, service status and competent accessibility or fire determinations govern; sensing cannot certify inclusion. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, comparison and recommendation. Technique: hard-constraint filtering and bounded recommendation. Decision interval: the person's stated decision interval. Minimum evidence: only the requirements voluntarily stated plus current whole-route and room evidence. Method: exclude candidates that fail a stated hard constraint, compare remaining options on declared preferences and show evidence status for every criterion. Uncertainty: personal variability, unverified support arrangements and rapidly changing asset status. Safe failure: return no recommendation and direct the person or planner to the responsible service when a critical criterion is unconfirmed; otherwise return not assessable.

ANSWER BOUNDARY

Do not demand a diagnosis, infer protected traits or severity, rank needs, store requirements beyond necessity, or certify that a space will meet an individual's needs.

WHY THIS IS USEFUL

Supports inclusive option finding from person-owned requirements while keeping professional and personal judgement in control.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

Stakeholder Question Catalogue 29 | Category 5: Accessibility, inclusion and diverse user needs

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5. Accessibility, inclusion and diverse user needs

SP-043 to SP-044 | 10 questions | Preserve choice, dignity and workable alternatives through consultation, current surveys and competent assessment.

Questions 3-4 of 10

Category 5 of 8

SP-043

LAYOUT BARRIER SCREEN

OUTPUT: Review priority

EVIDENCE: Measured survey + criteria

SPACE-PLANNING TEAMS QUESTION

Which current or proposed layouts require competent review because of clear-width, turning, reach, sightline, seating-choice or obstruction concerns?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: authorised measured geometry, furniture dimensions and optional item-position tags; no cameras or person-based movement capture. Spatial coverage: the room, entrances, circulation, furniture, controls, presentation points and accessible seating locations. Sampling and aggregation: a dated current-layout snapshot and scenario version. Quality gate: retain survey method, instrument calibration or validation, survey date, coordinate reference, tolerance, full boundary or route coverage, obscured areas, revision and competent provenance. A survey is a dated observation, not approval. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: accessibility/inclusion, room, capacity, fire-safety, facilities, AV and building-information owners. Records: current measured room and furniture geometry, approved layouts, clearances, reach/sightline criteria, assistive provision, egress requirements and user consultation evidence. Time basis: the current or proposed layout interval. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current access audits, user-stated requirements, service status and competent accessibility or fire determinations govern; sensing cannot certify inclusion. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: calculation and simulation. Technique: geometry-based barrier screening. Decision interval: the stated current or scenario layout. Minimum evidence: current measured geometry, exact layout and owner-defined criteria applicable to the activity. Method: calculate clearances, reach zones, sightlines and choice distribution; identify potential constraints and sensitivity to furniture position. Uncertainty: individual variation, temporary items, criteria requiring professional judgement and incomplete surveys. Safe failure: label results as potential barriers and require competent confirmation before any accessibility conclusion; otherwise return not assessable.

ANSWER BOUNDARY

Do not simulate a diagnosis, certify ergonomic or accessible compliance, move furniture, or decide that one nominal accessible position is sufficient.

WHY THIS IS USEFUL

Helps planners preserve genuine choice and identify layouts that need inclusive design input before adoption.

SP-044

INCLUSIVE PROVISION AND FALLBACK

OUTPUT: Provision resilience map

EVIDENCE: Owner records + status

SPACE-PLANNING TEAMS QUESTION

Are accessible room, route, toilet, lift, rest and support options distributed across relevant locations and times, with a credible fallback when one is unavailable?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: current owner-issued lift/door status and optional anonymous room availability totals; occupancy evidence cannot establish inclusive suitability. Spatial coverage: all eligible room classes, routes and operating periods in the portfolio scope. Sampling and aggregation: a dated inventory plus scenario or term-level availability windows. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: accessibility/inclusion, space, room-booking/timetable, facilities, lift/door, security and activity owners. Records: current room types, route/access features, layouts, assistive provision, eligible hours, bookings, closures, planned changes and contingency options. Time basis: the selected term, operating model or future scenario. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current access audits, user-stated requirements, service status and competent accessibility or fire determinations govern; sensing cannot certify inclusion. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, calculation and comparison. Technique: portfolio coverage and concentration calculation. Decision interval: the selected planning horizon. Minimum evidence: current inclusive-feature evidence, room eligibility and time availability across every compared class. Method: calculate the number and diversity of eligible options by time, floor and function; test loss of key rooms or routes and show unassessable gaps. Uncertainty: incomplete route evidence, changing bookings, user needs not captured by generic features and status outages. Safe failure: report feature coverage, not certified accessibility, and withhold claims for unverified routes; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer how many disabled people will use a space, treat one feature as universal suitability, or reduce inclusive provision because current observed use appears low.

WHY THIS IS USEFUL

Shows whether inclusive choice is distributed across the portfolio or depends on a small number of fragile rooms, routes or assets.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

Stakeholder Question Catalogue 29 | Category 5: Accessibility, inclusion and diverse user needs

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5. Accessibility, inclusion and diverse user needs

SP-045 to SP-046 | 10 questions | Preserve choice, dignity and workable alternatives through consultation, current surveys and competent assessment.

Questions 5-6 of 10

Category 5 of 8

SP-045ACCESSIBLE-JOURNEY MOVE IMPACTOUTPUT: Journey-impact comparisonEVIDENCE: Current surveys + scenario

SPACE-PLANNING TEAMS QUESTION

How would each move option change step-free journeys to entrances, toilets, lifts, support, welfare and shared facilities?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: current owner-issued route, lift and door status; optional anonymous route load used only as contextual crowding evidence. Spatial coverage: complete current and proposed journeys, including every transfer, doorway, lift, amenity and destination. Sampling and aggregation: status for the proposed operating interval; route load in one-minute aggregates across comparable periods. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: accessibility/inclusion, space, route, lift/door, facilities, security and activity owners. Records: current and proposed locations, whole-route access features, amenities, support locations, opening, closures, emergency dependencies and user-stated requirements from consultation. Time basis: the proposed move horizon and operating times. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current access audits, user-stated requirements, service status and competent accessibility or fire determinations govern; sensing cannot certify inclusion. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: simulation and comparison. Technique: whole-journey route comparison. Decision interval: the proposed post-move operating intervals. Minimum evidence: current and proposed locations, verified eligible routes and stated functional requirements or generic feature criteria. Method: compare distance, transfers, doors, vertical movement, route complexity, crowding context and fallback availability; keep each criterion separate. Uncertainty: changing asset status, individual variation, temporary obstructions and incomplete consultation. Safe failure: state which journey elements cannot be compared and do not declare the move inclusive; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer a person's mobility or tolerance, promise an equivalent journey, certify accessibility, or decide that extra distance is acceptable.

WHY THIS IS USEFUL

Makes inclusive journey consequences visible before a move is approved or communicated.

SP-046SENSORY CONDITIONS AND CHOICEOUTPUT: Condition-choice profileEVIDENCE: Environmental telemetry

SPACE-PLANNING TEAMS QUESTION

Which rooms provide stable or variable sound, light, temperature and air conditions relevant to user-stated preferences, and where is meaningful choice limited?

SENSORS AND TELEMETRY

Proposed modality and location: room-level temperature, relative humidity, CO2, illuminance and numerical acoustic indicators; no audio recording or emotion/health sensing. Spatial coverage: each named candidate room and occupied zone, with no corridor substitution; daylight and local control zones mapped explicitly. Sampling and aggregation: one-minute observations aggregated to five- or 15-minute distributions over representative weeks and the requested forecast window. Quality gate: document sensor height and representative placement, units and calibration or validation; align clocks within 60 seconds; report cadence, spatial variation, uptime, gaps, saturation, drift, maintenance and processing lineage. Readings describe sampled conditions only. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, BMS/facilities, accessibility/inclusion, room, booking/timetable, weather and sensor-data owners. Records: current room purpose, controls, opening, bookings, known equipment or envelope issues, weather and any user-stated functional preferences. Time basis: the representative baseline and proposed use interval. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current access audits, user-stated requirements, service status and competent accessibility or fire determinations govern; sensing cannot certify inclusion. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation and comparison. Technique: variability comparison and bounded forecast. Decision interval: the requested use interval. Minimum evidence: valid room-level environmental history, current controls/status and user-stated criteria where supplied. Method: calculate distributions, rate of change and forecast ranges from matched conditions; rank only eligible rooms and report criterion-level uncertainty. Uncertainty: personal response, local seat variation, future weather/control changes and sparse comparable conditions. Safe failure: provide measured ranges and alternatives rather than asserting comfort or suitability; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer neurotype, disability, distress, comfort, health effect or protected trait, and do not certify that a space is low-stimulation or suitable for an individual.

WHY THIS IS USEFUL

Allows planners to preserve a range of environmental choices and identify where room-level evidence is missing.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

5. Accessibility, inclusion and diverse user needs

Questions 7-8 of 10

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SP-047 to SP-048 | 10 questions | Preserve choice, dignity and workable alternatives through consultation, current surveys and competent assessment.

SP-047

REST, PRIVACY AND LOW-STIMULATION PROVISION

OUTPUT: Provision gap map

EVIDENCE: Owner records + status

SPACE-PLANNING TEAMS QUESTION

Where are current designated rest, privacy, wellbeing or low-stimulation spaces, what hours are they usable, and which planning zones lack a credible alternative?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional anonymous aggregate availability or room-status evidence; no sensing of need, distress, health or individual use. Spatial coverage: only owner- designated spaces and eligible routes, with small sensitive locations disclosed according to permission. Sampling and aggregation: current availability status and a term-level provision snapshot. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: accessibility/inclusion, well-being, space, room-booking, facilities, security and local service owners. Records: current designated provision, purpose, access and booking rules, opening, amenities, support contacts, closures, confidentiality and planned changes. Time basis: the current operating model and stated future scenario. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current access audits, user-stated requirements, service status and competent accessibility or fire determinations govern; sensing cannot certify inclusion. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: provision inventory and gap classification. Decision interval: the current or scenario horizon. Minimum evidence: owner-confirmed designated spaces, eligible routes, availability rules and user consultation criteria. Method: map provision by type, time and location, test resilience to closures and classify gaps that need consultation rather than data inference. Uncertainty: informal spaces, variable personal needs and sensitive information that cannot be widely disclosed. Safe failure: state only confirmed provision and route unresolved needs to the responsible service; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer who needs or uses these spaces, disclose sensitive locations beyond permission, diagnose distress, or label an ordinary quiet room as designated provision.

WHY THIS IS USEFUL

Supports equitable distribution of person-owned options without turning sensitive support spaces into monitoring targets.

SP-048

INCLUSIVE EMERGENCY DEPENDENCIES

OUTPUT: Competent-review triggers

EVIDENCE: Safety records + scenario

SPACE-PLANNING TEAMS QUESTION

Which space options alter step-free egress, refuge access, evacuation-equipment locations or assistance arrangements and therefore require competent assessment?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: owner-issued life-safety, lift or door status may be displayed only as current authoritative context; generic sensors cannot validate an evacuation strategy. Spatial coverage: the affected spaces, complete emergency routes, refuges, equipment and assistance points. Sampling and aggregation: current status and the proposed scenario interval, with event timestamps and explicit system health. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: fire-safety, emergency, accessibility/inclusion, space, security, facilities and building-information owners. Records: current fire strategy, evacuation routes, refuges, equipment, assistance arrangements, capacities, impairments, drills/exercises evidence and proposed spatial change. Time basis: the scenario horizon and current emergency arrangements. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current access audits, user-stated requirements, service status and competent accessibility or fire determinations govern; sensing cannot certify inclusion. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and simulation. Technique: authoritative dependency mapping and scenario impact classification. Decision interval: the proposed spatial scenario. Minimum evidence: current competent-owner emergency strategy and exact scenario geometry/use assumptions. Method: map affected routes, capacities and assistance dependencies; classify unchanged, potentially affected, unresolved and outside the service's competence. Uncertainty: emergency conditions, person-specific plans, temporary impairments and professional interpretation. Safe failure: stop at impact identification and refer every potentially affected element to competent fire/emergency/accessibility owners; otherwise return not assessable.

ANSWER BOUNDARY

Do not certify evacuation safety, create or disclose personal emergency plans, direct emergency action, operate life-safety systems or approve the scenario.

WHY THIS IS USEFUL

Ensures that apparently ordinary space changes expose their inclusive emergency dependencies early.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

5. Accessibility, inclusion and diverse user needs

Questions 9-10 of 10

Category 5 of 8

SP-049 to SP-050 | 10 questions | Preserve choice, dignity and workable alternatives through consultation, current surveys and competent assessment.

SP-049

INCLUSION CONSULTATION AND EVIDENCE GAPS

OUTPUT: Consultation brief

EVIDENCE: Consultation + survey

SPACE-PLANNING TEAMS QUESTION

Which functional requirements and lived-experience questions remain unresolved, and what accessible consultation or measured survey is needed before options are compared?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: quality and coverage metadata for any proposed route, room or environmental evidence; no additional person monitoring. Spatial coverage: the exact requirement and complete journey or space to which the claim would apply. Sampling and aggregation: assessed for the current decision and proposed operating interval. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: the person or user group through authorised consultation, accessibility/inclusion, space, facilities, route, room and relevant professional owners. Records: current access and room evidence, consultation findings, measured surveys, assistive provision, policies and known evidence gaps. Time basis: the specific planning decision and scenario. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current access audits, user-stated requirements, service status and competent accessibility or fire determinations govern; sensing cannot certify inclusion. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, comparison and recommendation. Technique: claim-to-evidence answerability classification. Decision interval: the current planning decision. Minimum evidence: a precise functional question and the full set of current relevant records and evidence limits. Method: classify the question as owner-confirmed, measurable with proportionate data, consultation-dependent, competent-survey-dependent or not answerable. Uncertainty: individual variation, emerging requirements, incomplete consultation and evidence that covers only part of a journey. Safe failure: state the missing evidence and responsible route rather than generating an inferred answer; otherwise return not assessable.

ANSWER BOUNDARY

Do not invent user needs, substitute averages for lived experience, demand disclosure of diagnoses, or present data availability as evidence of inclusion.

WHY THIS IS USEFUL

Prevents fluent but unsupported inclusion claims and directs planners to the right human or professional source.

SP-050

INCLUSION TRADE-OFF COMPARISON

OUTPUT: Qualified option comparison

EVIDENCE: Criteria + consultation

SPACE-PLANNING TEAMS QUESTION

Among options that pass every hard constraint, which preserve the broadest range of stated needs, dignity, choice and workable fallback arrangements?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: current route/asset status and room-level environmental summaries only where directly relevant to voluntarily stated requirements. Spatial coverage: every candidate space, complete route, alternative facility and operating time in each scenario. Sampling and aggregation: current baseline plus bounded future scenarios; environmental data in five- or 15-minute aggregates. Quality gate: verify asset identity, room or service mapping and state semantics through commissioning or stocktake; retain event time, stale-state timeout, expected heartbeat, uptime, gaps, relocations, duplicates, firmware and source lineage. A heartbeat is not proof of suitability. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: accessibility/inclusion, space, room, route, facilities, activity, booking and user-consultation owners. Records: scenario layouts, route/access features, capacities, amenities, assistive provision, availability, closures and person- or group-stated functional requirements collected with permission. Time basis: the stated scenario horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current access audits, user-stated requirements, service status and competent accessibility or fire determinations govern; sensing cannot certify inclusion. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: comparison and recommendation. Technique: constraint filtering and Pareto comparison. Decision interval: the scenario horizon. Minimum evidence: feasible scenarios, complete current evidence and explicit functional criteria with importance stated by users or owners. Method: exclude failed hard constraints, compare remaining scenarios without collapsing diverse needs into one score, and show trade-offs and options lost. Uncertainty: unrepresented requirements, future status changes and values that cannot be legitimately weighted by the service. Safe failure: return a non-dominated option set and unresolved consultation questions rather than one automatic winner; otherwise return not assessable.

ANSWER BOUNDARY

Do not rank people or needs, infer protected traits, assign hidden weights, certify an inclusive outcome, or choose the final scenario.

WHY THIS IS USEFUL

Gives decision-makers transparent inclusive trade-offs while preserving consultation and accountable judgement.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

Stakeholder Question Catalogue 29 | Category 5: Accessibility, inclusion and diverse user needs

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6. Moves, scenarios, decants and change impacts

Questions 1-2 of 10

Category 6 of 8

SP-051 to SP-052 | 10 questions | Compare moves and temporary arrangements with dependencies, disruption, consultation, reversibility and named decision gates.

SP-051

MOVE IMPACT REGISTER

OUTPUT: Impact register

EVIDENCE: Owner records

SPACE-PLANNING TEAMS QUESTION

Which spaces, functions, assets, bookings, storage, services, adjacencies, routes and support arrangements would this move affect?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: content-free asset/service status and optional authorised equipment location tags; no monitoring of staff, research activity or belongings. Spatial coverage: the current space, candidate destination, affected shared areas, routes and directly serving assets. Sampling and aggregation: a dated current snapshot and the proposed move/occupation windows. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space and activity owners, moves/logistics, facilities, IT/AV, M&E; accessibility, security, safety, storage and booking owners. Records: current functions, space boundaries, assets, storage, services, access, adjacencies, bookings, restrictions, dependencies, licences and planned changes. Time basis: the proposed move and post-move operating horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current project, service, booking, access, funding and change-control records govern; a scenario or observed pattern does not authorise a move. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: current-to-proposed dependency comparison. Decision interval: the move and stabilisation horizon. Minimum evidence: verified origin and destination, owner-confirmed function requirements and complete dependency inventories. Method: map each current dependency to an equivalent, changed, missing or unknown destination provision and identify upstream impacts. Uncertainty: undocumented local equipment, informal storage, evolving work patterns and future service changes. Safe failure: provide a gap list and stop before declaring fit when any critical dependency is unresolved; otherwise return not assessable.

ANSWER BOUNDARY

Do not inspect research or work content, identify occupants, decide who moves, promise continuity, or authorise possession, access, disconnection or installation.

WHY THIS IS USEFUL

Creates a complete move-impact picture before dates, costs or destinations are treated as viable.

SP-052

DESTINATION READINESS

OUTPUT: Destination gate

EVIDENCE: Records + surveys

SPACE-PLANNING TEAMS QUESTION

Which candidate destinations pass every hard requirement now, which need defined enabling work, and which cannot yet be assessed?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: content-free room/asset status, route/lift/door state and optional environmental baselines relevant to stated requirements. Spatial coverage: all candidate rooms, full move and user routes, shared facilities and directly serving assets. Sampling and aggregation: current status plus the planned decant and operating intervals; environmental data in 15-minute representative summaries. Quality gate: verify asset identity, room or service mapping and state semantics through commissioning or stocktake; retain event time, stale-state timeout, expected heartbeat, uptime, gaps, relocations, duplicates, firmware and source lineage. A heartbeat is not proof of suitability. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space and activity owners, moves/logistics, room/booking, capacity, accessibility, facilities, IT/AV, M&E; security and safety owners. Records: move requirement set, current candidate inventory, capacity/layout, equipment/services, routes, access, bookings, closures, enabling works and future commitments. Time basis: the proposed move and occupation intervals. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current project, service, booking, access, funding and change-control records govern; a scenario or observed pattern does not authorise a move. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, comparison and recommendation. Technique: hard-constraint filtering and scenario comparison. Decision interval: the move plus intended occupation period. Minimum evidence: an owner-approved requirement set and current authoritative evidence for every candidate and route. Method: exclude failed candidates, compare remaining options on declared preferences, show constraint margins and expose all enabling dependencies. Uncertainty: future asset status, hidden local requirements, temporary closures and incomplete inclusive evidence. Safe failure: return no eligible destination where any critical constraint is missing or conflicting; otherwise return not assessable.

ANSWER BOUNDARY

Do not assign a destination, relax hard constraints, certify suitability, promise availability, or authorise access, works or the move.

WHY THIS IS USEFUL

Produces a defensible feasible set for accountable owners rather than a ranked list built on incomplete assumptions.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

6. Moves, scenarios, decants and change impacts

SP-053 to SP-054 | 10 questions | Compare moves and temporary arrangements with dependencies, disruption, consultation, reversibility and named decision gates.

Questions 3-4 of 10

Category 6 of 8

SP-053

MOVE SEQUENCING

OUTPUT: Sequenced move plan

EVIDENCE: Project scenario

SPACE-PLANNING TEAMS QUESTION

Which feasible move sequence best protects service continuity, accessible routes, booked commitments and technical dependencies within approved change windows?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: owner-issued room, asset, lift, door and network/service status; optional non-personal move-task completion events. Spatial coverage: origin, destination, temporary holding areas, routes and every dependent asset/service. Sampling and aggregation: event-driven status across the approved preparation, move and stabilisation windows. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: moves/logistics, space, activity, facilities, IT/AV, M&E; security/access, booking/timetable, safety and contractor owners. Records: approved windows, task dependencies, room possession, bookings, access/permit constraints, service cutover plans, setup tests, handover criteria and contingency options. Time basis: the authorised move horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current project, service, booking, access, funding and change-control records govern; a scenario or observed pattern does not authorise a move. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: simulation, optimisation and recommendation. Technique: precedence-constrained schedule optimisation. Decision interval: the approved preparation, move and stabilisation windows. Minimum evidence: owner-approved tasks, durations/ranges, dependencies, eligible windows and resource constraints. Method: construct a dependency network, generate feasible sequences, minimise stated disruption metrics and stress-test delays or failed tasks. Uncertainty: task-duration variability, late access changes, hidden dependencies and owner approvals not yet granted. Safe failure: show only conditional sequences and identify the first owner decision required; never present a draft as authority; otherwise return not assessable.

ANSWER BOUNDARY

Do not issue work, instruct contractors, grant access, approve permits, disconnect services, control assets or promise a completion date.

WHY THIS IS USEFUL

Helps move coordinators identify feasible sequencing and contingency points before formal plans are issued.

SP-054

DECANT SUITABILITY

OUTPUT: Decant gate

EVIDENCE: Records + observation

SPACE-PLANNING TEAMS QUESTION

Can each proposed decant space support the required activity, peak demand, equipment, storage, routes, environment and user support for the full temporary period?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous aggregate use and room-level environmental or content-free service evidence where relevant; no monitoring of work content or individual output. Spatial coverage: candidate decant rooms, routes, shared facilities and directly serving assets. Sampling and aggregation: 15-minute representative baselines plus current status for the full decant interval and peak periods. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space and activity owners, booking/timetable, capacity, accessibility, facilities, IT/AV, M&E; security, safety and project owners. Records: temporary activity requirements, decant duration, candidate eligibility, layouts, capacity, services, access, bookings, closures, support arrangements and planned changes. Time basis: the complete decant and recovery period. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current project, service, booking, access, funding and change-control records govern; a scenario or observed pattern does not authorise a move. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: estimate, simulation and comparison. Technique: temporary-fit and peak-demand simulation. Decision interval: the complete decant period. Minimum evidence: owner-approved requirements, valid candidate evidence and peak/seasonal demand assumptions. Method: test hard constraints for every phase, simulate concurrent demand and service load, identify periods with no feasible capacity and show sensitivity. Uncertainty: long-duration changes, demand adaptation, asset reliability and temporary arrangements not yet confirmed. Safe failure: state that decant capability is unconfirmed where any critical period or dependency lacks evidence; otherwise return not assessable.

ANSWER BOUNDARY

Do not certify temporary accommodation, promise business continuity, infer staff performance, or approve occupancy, access, services or works.

WHY THIS IS USEFUL

Prevents a short-term decant from being accepted on average capacity while failing at peaks or later phases.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

6. Moves, scenarios, decants and change impacts

SP-055 to SP-056 | 10 questions | Compare moves and temporary arrangements with dependencies, disruption, consultation, reversibility and named decision gates.

Questions 5-6 of 10

Category 6 of 8

SP-055

TEMPORARY CAPACITY LOSS

OUTPUT: Disruption forecast

EVIDENCE: Programme + demand records

SPACE-PLANNING TEAMS QUESTION

How much usable capacity and room-type supply will be unavailable during works, and where could displaced demand exceed supported alternatives?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: owner-issued room/service closure status and optional anonymous aggregate historical use for calibration. Spatial coverage: the affected rooms, room classes, authorised alternatives and shared facilities across the portfolio scope. Sampling and aggregation: the exact closure phases, with historical use in comparable 15-minute or session-level aggregates. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: project/works, space, room-booking/timetable, capacity, accessibility, facilities and activity owners. Records: phased closure plan, current room inventory, capacity by layout, eligible hours, bookings, known future demand, alternative supply and concurrent works. Time basis: each closure phase and recovery window. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current project, service, booking, access, funding and change-control records govern; a scenario or observed pattern does not authorise a move. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: calculation, estimate and simulation. Technique: capacity-loss calculation and demand redistribution simulation. Decision interval: each stated closure phase. Minimum evidence: authoritative closure scope, current room supply and owner-approved demand assumptions. Method: remove affected supply by room class and time, re-test demand against eligible alternatives, identify bottlenecks and report unmet ranges. Uncertainty: demand changes, rooms with uncertain future availability and informal substitution not represented in records. Safe failure: provide bounded impacts and list unresolved supply rather than assuming inappropriate alternatives; otherwise return not assessable.

ANSWER BOUNDARY

Do not declare a real closure, assign displaced activities, relax capacity or suitability, or approve the works programme.

WHY THIS IS USEFUL

Shows whether planned loss can be absorbed or needs advance timetable, event or decant intervention.

SP-056

ENABLING WORKS AND DECISIONS

OUTPUT: Precondition schedule

EVIDENCE: Project + owner records

SPACE-PLANNING TEAMS QUESTION

Which surveys, enabling works, permissions, shutdowns, commissioning checks and accountable decisions must be complete before the move starts?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: content-free asset/service status, owner-issued task completion and commissioning events; generic sensing cannot prove completion. Spatial coverage: the destination, movement route, affected services and each precondition named by an owner. Sampling and aggregation: event-driven evidence through the pre-move window, with signed or owner-issued status timestamps. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: project, facilities, M&E, IT/AV, space, moves/logistics, security/access, safety, accessibility, contractor and commissioning owners. Records: scope, surveys, designs, permits, access, service modifications, equipment installation, testing, cleaning, furniture setup, handover and owner acceptance criteria. Time basis: the preparation and go/no-go horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current project, service, booking, access, funding and change-control records govern; a scenario or observed pattern does not authorise a move. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: precondition and dependency classification. Decision interval: the preparation window. Minimum evidence: the complete owner-issued precondition list, current task status and acceptance evidence. Method: classify each dependency as satisfied, scheduled, blocked, expired, conflicting or outside the service's authority and identify critical-path consequences. Uncertainty: late discoveries, unsigned acceptance, scope changes and status feeds that lag physical work. Safe failure: treat uncertain or owner-unaccepted work as incomplete and direct it to the responsible role; otherwise return not assessable.

ANSWER BOUNDARY

Do not approve surveys, designs, permits, testing, commissioning or handover, and do not authorise contractors or operate services.

WHY THIS IS USEFUL

Makes hidden prerequisites and accountable owners visible before a move date is committed.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

Stakeholder Question Catalogue 29 | Category 6: Moves, scenarios, decants and change impacts

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6. Moves, scenarios, decants and change impacts

SP-057 to SP-058 | 10 questions | Compare moves and temporary arrangements with dependencies, disruption, consultation, reversibility and named decision gates.

Questions 7-8 of 10

Category 6 of 8

SP-057

POST-MOVE NETWORK AND AMENITY IMPACT

OUTPUT: Network-impact comparison

EVIDENCE: Governed scenario

SPACE-PLANNING TEAMS QUESTION

How would each move option change aggregate journey distance, route pressure and demand on lifts, toilets, kitchens, storage, meeting and support space?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous area-level counts and route totals with no identity linkage; optional shared-facility demand and environmental indicators. Spatial coverage: current and proposed origin zones, eligible routes, shared facilities and affected destination areas. Sampling and aggregation: 15-minute representative baselines projected to the stated operating schedule. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space and activity owners, timetable/booking, accessibility, route/lift/door, facilities, security and shared- service owners. Records: current and proposed locations, authorised aggregate schedules/demand, routes, access, shared-facility capacities, closures and alternative provisions. Time basis: the proposed post-move operating model. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current project, service, booking, access, funding and change-control records govern; a scenario or observed pattern does not authorise a move. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: estimate, simulation and comparison. Technique: aggregate route and shared-service demand simulation. Decision interval: the proposed post-move horizon. Minimum evidence: verified current baseline, complete scenario placement and owner-approved aggregate demand assumptions. Method: redistribute demand by time, calculate route-distance and load changes, test shared-service pressure and compare feasible alternatives. Uncertainty: behavioural adaptation, remote work, unscheduled activity and incomplete origin-zone evidence. Safe failure: show ranges and unmodelled demand; do not claim observed future journeys; otherwise return not assessable.

ANSWER BOUNDARY

Do not track or predict individual journeys, infer collaboration or satisfaction, certify route safety/accessibility, or decide the move.

WHY THIS IS USEFUL

Exposes building-wide consequences that are invisible in a destination-only space fit assessment.

SP-058

CONSULTATION, NOTICE AND LEARNING

OUTPUT: Engagement and learning plan

EVIDENCE: Consultation + evidence

SPACE-PLANNING TEAMS QUESTION

Which affected user groups and service owners need accessible consultation or notice, and what before-and-after evidence will test the move assumptions?

SENSORS AND TELEMETRY

The core evidence is consultation and project records. Proposed corroboration, only if proportionate and agreed, may use anonymous room or route counts, numerical environmental summaries and content-free service status before and after the change. Spatial coverage: affected spaces, routes and shared facilities only. Timing and aggregation: a declared baseline and follow-up window, aggregated to at least 15 minutes with no identity linkage; voluntary feedback remains separate and access-controlled. Quality gate: document sensor height and representative placement, units and calibration or validation; align clocks within 60 seconds; report cadence, spatial variation, uptime, gaps, saturation, drift, maintenance and processing lineage. Readings describe sampled conditions only. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: the project or change lead, local space owners, accessibility and inclusion lead, service owners, communications lead, consultation owner and data steward. Records: stakeholder map by role or affected group, consultation plan, notices, agreed requirements, decisions, issues, baseline assumptions, outcome measures and retention or permission terms. Time basis: consultation, notice, move and follow-up dates. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current project, service, booking, access, funding and change-control records govern; a scenario or observed pattern does not authorise a move. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, comparison and recommendation. Decision interval: from option development through the agreed post-change review. Minimum evidence: affected functions and groups, accountable owners, an accessible engagement route, notice requirements, baseline assumptions and approved outcome measures. Method: map who must be consulted, informed or asked to validate; compare before and after aggregate indicators only against predeclared assumptions; record dissent and non-response without treating either as consent. Uncertainty: incomplete participation, adaptation time and concurrent changes. Safe failure: issue an engagement and evidence gap list and defer conclusions until the minimum consultation is met.

ANSWER BOUNDARY

Do not identify individuals in reporting, infer consent from silence, require sensitive disclosure, turn voluntary feedback into surveillance, or claim that before-and-after association proves the move caused an outcome. Accountable owners decide whether to proceed, adapt or reverse.

WHY THIS IS USEFUL

Builds consultation, notice and learning into the move plan so operational disruption and unmet needs are found early.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

6. Moves, scenarios, decants and change impacts

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SP-059 to SP-060 | 10 questions | Compare moves and temporary arrangements with dependencies, disruption, consultation, reversibility and named decision gates.

SP-059

REVERSIBLE TRIAL AND TRIGGERS

OUTPUT: Trial-and-trigger plan

EVIDENCE: Governed scenario

SPACE-PLANNING TEAMS QUESTION

Which parts of the proposed change can be trialled or reversed, what would reversal cost or disrupt, and what evidence thresholds trigger pause, adaptation or rollback?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: owner-issued task, asset and room status plus only the aggregate outcome indicators explicitly approved for the scenario. Spatial coverage: the changed spaces, retained alternatives, affected services and agreed evidence points. Sampling and aggregation: from baseline through stabilisation and the defined decision checkpoints. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space decision owner, project/moves, finance, facilities, activity, accessibility, security and relevant service owners. Records: scenario decision record, irreversible commitments, leases/contracts, retained assets, reinstatement needs, approved indicators, thresholds, consultation commitments and decision dates. Time basis: the whole scenario lifecycle and stated checkpoints. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current project, service, booking, access, funding and change-control records govern; a scenario or observed pattern does not authorise a move. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: calculation, comparison and recommendation. Technique: reversibility classification and trigger sensitivity. Decision interval: the stated decision checkpoints. Minimum evidence: current contractual/physical constraints, baseline evidence and owner-approved trigger definitions. Method: classify decisions as readily reversible, conditionally reversible or effectively irreversible; calculate evidence ranges around triggers and test false-alarm risk. Uncertainty: lagging outcomes, ambiguous causality, sunk costs and changing organisational priorities. Safe failure: present trigger status as decision support only and refer every action to the accountable owner; otherwise return not assessable.

ANSWER BOUNDARY

Do not reverse a move, cancel commitments, promise reversibility, or use a single metric to override consultation, safety, accessibility or contractual constraints.

WHY THIS IS USEFUL

Helps planners design cautious, learnable change rather than locking into assumptions that cannot be tested or undone.

SP-060

MOVE DECISION-READINESS GATE

OUTPUT: Named blocker list

EVIDENCE: Decision records

SPACE-PLANNING TEAMS QUESTION

Which evidence, permission, funding decision, consultation outcome or accountable-owner approval still blocks this move, decant or space-change option?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: source-health and mapping metadata for any proposed evidence; no new sensing is required to classify decision gaps. Spatial coverage: the complete scenario boundary, affected rooms/routes/services and every required evidence source. Sampling and aggregation: current at the decision checkpoint, with effective and retrieval times preserved. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: the accountable space decision owner and all capacity, accessibility, fire/safety, facilities, security, finance, project, activity, booking and information owners touched by the scenario. Records: current scenario, constraints, surveys, consents, costs, demand evidence, consultation inputs, risks, dependencies and owner decisions. Time basis: the stated decision checkpoint. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current project, service, booking, access, funding and change-control records govern; a scenario or observed pattern does not authorise a move. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: decision-readiness and blocker classification. Decision interval: the stated checkpoint. Minimum evidence: a complete decision requirement matrix and current source status for every criterion. Method: classify each item as confirmed, conditional, missing, conflicting, expired, permission-restricted or outside the service's competence; identify dependency order. Uncertainty: requirements discovered late, unresolved authority and evidence that may change before action. Safe failure: provide only a readiness map and named next evidence owner; do not produce an approval conclusion; otherwise return not assessable.

ANSWER BOUNDARY

Do not approve, reject or prioritise the option, conceal unresolved conditions, or treat a complete data table as proof that professional and consultation duties are complete.

WHY THIS IS USEFUL

Gives accountable decision-makers a clear, traceable list of what is known, conditional and still unresolved.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

7. Teaching, research, workplace and event demand

SP-061 to SP-062 | 10 questions | Translate teaching, research, workplace and event requirements into time-specific demand without judging people or work.

Questions 1-2 of 10

Category 7 of 8

SP-061

TEACHING DEMAND BY ROOM CLASS

OUTPUT: Room-class forecast

EVIDENCE: Timetable + scenario

SPACE-PLANNING TEAMS QUESTION

What teaching-space demand is expected by capacity band, layout, equipment, accessibility requirement and timetable period under the approved planning assumptions?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional anonymous historical room counts may calibrate realised-use ranges but cannot replace enrolment, timetable or capacity records. Spatial coverage: teaching rooms with equivalent room-level coverage, grouped only after current room-class and layout verification. Sampling and aggregation: session-level or 15-minute aggregates across comparable teaching weeks; no named-student data. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: planning/enrolment, programme, timetabling, room, capacity, accessibility, AV/IT and space owners. Records: approved enrolment ranges, activity patterns, group-size rules, draft timetable, room classes, capacity by layout, equipment/access requirements, closures and planned changes. Time basis: the future timetable cycle and approved scenarios. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current timetable, programme, research, workplace, event and room-owner records govern; aggregate observations cannot judge attendance, performance or activity value. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: calculation and forecast. Technique: aggregate demand calculation and scenario forecast. Decision interval: the future timetable cycle. Minimum evidence: approved demand scenarios, activity requirements and current room supply by eligible class. Method: sum concurrent demand by room class and time, test lower/central/upper scenarios, separate scheduled demand from optional observed-use calibration and report unmet ranges. Uncertainty: late enrolment, curriculum change, room outages and behavioural variation. Safe failure: show bounded demand and data gaps; do not invent supply or relax activity constraints; otherwise return not assessable.

ANSWER BOUNDARY

Do not predict named student attendance, change group sizes, assign rooms, publish a timetable or certify capacity.

WHY THIS IS USEFUL

Shows which teaching-space classes create structural pressure before allocations are finalised.

SP-062

TIMETABLE-ROOM MISMATCH

OUTPUT: Mismatch action queue

EVIDENCE: Timetable + room records

SPACE-PLANNING TEAMS QUESTION

Where do scheduled teaching activities conflict with the approved capacity, layout, services, equipment or access conditions of the allocated room?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional anonymous occupancy bands and content-free AV/service health may provide separate realised-use context. Spatial coverage: the allocated room and its directly serving equipment/services; no inspection of teaching content or attendance identity. Sampling and aggregation: session-level comparison across the selected timetable cycle, with current status at each session. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: programme/activity, timetabling, room, capacity, accessibility, AV/IT and space owners. Records: current activity requirements, allocated rooms, capacity by layout, equipment, access, setup, linked spaces, substitutions, cancellations and exceptions. Time basis: the selected timetable cycle. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current timetable, programme, research, workplace, event and room-owner records govern; aggregate observations cannot judge attendance, performance or activity value. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: activity-requirement versus allocation comparison. Decision interval: the selected timetable cycle. Minimum evidence: owner-approved requirements and current room constraints for every allocated session. Method: classify exact fit, constrained fit, over-provision, under-provision, exception and unknown; show whether alternatives were actually eligible at that time. Uncertainty: requirements recorded too broadly, late room changes and equipment health not captured in static inventories. Safe failure: withhold a mismatch label when the requirement or current layout is unverified; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer teaching quality, engagement or attendance, criticise named staff, reallocate sessions or treat a larger room as waste without context.

WHY THIS IS USEFUL

Identifies room-class imbalances and recording gaps that can inform future supply and timetable design.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

7. Teaching, research, workplace and event demand

Questions 3-4 of 10
Category 7 of 8

SP-063 to SP-064 | 10 questions | Translate teaching, research, workplace and event requirements into time-specific demand without judging people or work.

SP-063

RESEARCH DEMAND ENVELOPE

OUTPUT: Demand envelope

EVIDENCE: Programme + asset records

SPACE-PLANNING TEAMS QUESTION

What aggregate space, storage, equipment, environmental-control and collaboration demand is approved or credibly forecast for each research-space class?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: only content-free asset/service utilisation totals and anonymous area-level use where authorised; no inspection of research data, experiments, devices or communications. Spatial coverage: research-space classes and shared infrastructure, aggregated enough to protect sensitive projects and small groups. Sampling and aggregation: monthly or project-phase bands over the approved planning horizon; no individual timelines. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: research leadership, project and laboratory owners, space, finance, facilities, M&E; IT, safety, security and information-governance owners. Records: approved project portfolios, generic activity requirements, grants or commitments, equipment/storage/service needs, licences, hazards, planned starts/ends and current allocations at an authorised aggregate level. Time basis: the approved research-planning horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current timetable, programme, research, workplace, event and room-owner records govern; aggregate observations cannot judge attendance, performance or activity value. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: calculation, estimate and forecast. Technique: aggregate requirement synthesis and scenario range. Decision interval: the approved research-planning horizon. Minimum evidence: permissioned project-level requirements aggregated to safe categories plus current supply and committed changes. Method: group requirements by space/service class, time and certainty; calculate committed, probable and contingent ranges without exposing project content. Uncertainty: funding outcomes, project changes, confidential requirements and demand not captured centrally. Safe failure: suppress sensitive cells and label incomplete categories rather than infer hidden research demand; otherwise return not assessable.

ANSWER BOUNDARY

Do not inspect research content, infer productivity or project value, disclose sensitive locations or plans, or decide which research receives space.

WHY THIS IS USEFUL

Supports forward space capacity planning while preserving research confidentiality and accountable prioritisation.

SP-064

RESEARCH INFRASTRUCTURE DEPENDENCIES

OUTPUT: Dependency graph

EVIDENCE: Technical records + consultation

SPACE-PLANNING TEAMS QUESTION

Which research functions depend on proximity to specialist infrastructure, controlled environments, secure storage, loading, authorised routes or technical support?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: content-free asset status, service- zone metadata and optional authorised equipment-location tags; no experiment, sample or network-content inspection. Spatial coverage: the relevant space classes, infrastructure, storage and approved routes, aggregated where disclosure is restricted. Sampling and aggregation: current topology plus the approved project horizon and planned service changes. Quality gate: verify asset identity, room or service mapping and state semantics through commissioning or stocktake; retain event time, stale-state timeout, expected heartbeat, uptime, gaps, relocations, duplicates, firmware and source lineage. A heartbeat is not proof of suitability. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: research/laboratory, space, facilities, M&E; IT, security, safety, storage and information owners. Records: generic requirement classes, infrastructure and service capacities, route/security constraints, storage/handling needs, licences, planned equipment and approved future projects. Time basis: the stated planning horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current timetable, programme, research, workplace, event and room-owner records govern; aggregate observations cannot judge attendance, performance or activity value. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: dependency-graph and proximity-constraint analysis. Decision interval: the stated research-planning horizon. Minimum evidence: owner-confirmed generic dependencies, current topology and scenario locations. Method: represent hard service, route and proximity constraints, identify clusters and single points of dependency, and test candidate placements without exposing sensitive details. Uncertainty: future equipment, confidential constraints, capacity under load and requirements known only locally. Safe failure: show a restricted or aggregated result and refer unresolved sensitive dependencies to authorised owners; otherwise return not assessable.

ANSWER BOUNDARY

Do not reveal sensitive research functions, infer project activity from asset status, rank projects, or authorise access, relocation or service connection.

WHY THIS IS USEFUL

Prevents space scenarios from separating research functions from infrastructure that cannot be duplicated or moved easily.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

7. Teaching, research, workplace and event demand

SP-065 to SP-066 | 10 questions | Translate teaching, research, workplace and event requirements into time-specific demand without judging people or work.

Questions 5-6 of 10

Category 7 of 8

SP-065

WORKPLACE SETTING MIX

OUTPUT: Setting-mix scenario

EVIDENCE: Consultation + aggregate evidence

SPACE-PLANNING TEAMS QUESTION

What mix of desks, focus rooms, collaboration areas, meeting rooms, storage and support space is justified by approved operating assumptions and consultation?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous area-level presence and use-duration bands, room/desk-group availability totals and numerical environmental indicators where proportionate. Spatial coverage: workplace zones and shared facilities, aggregated to prevent individual or team movement histories. Sampling and aggregation: 15-minute bands across representative weeks, with day types and remote/on-site policy scenarios separated. Quality gate: document sensor height and representative placement, units and calibration or validation; align clocks within 60 seconds; report cadence, spatial variation, uptime, gaps, saturation, drift, maintenance and processing lineage. Readings describe sampled conditions only. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: workplace/HR policy owner, space and local unit owners, room booking, accessibility/inclusion, facilities and IT owners. Records: approved workplace model, staffing ranges at aggregate level, role-independent activity requirements, attendance policies, existing provision, bookings, storage/support needs and consultation evidence. Time basis: the future workplace scenario horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current timetable, programme, research, workplace, event and room-owner records govern; aggregate observations cannot judge attendance, performance or activity value. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: calculation, estimate and simulation. Technique: aggregate requirement synthesis and scenario comparison. Decision interval: the approved workplace horizon. Minimum evidence: owner-approved operating assumptions, current provision and quality-controlled aggregate patterns plus consultation inputs. Method: calculate demand ranges by space type and time, preserve quiet/collaboration/meeting/support choices and test alternative mixes without maximising desk density alone. Uncertainty: policy change, self-selection, unrecorded visitors, individual needs and adaptation after change. Safe failure: show ranges and consultation gaps; do not infer a single optimal workplace ratio; otherwise return not assessable.

ANSWER BOUNDARY

Do not identify attendance, infer productivity, performance, collaboration quality or reasons for remote work, monitor devices, or assign individuals to workplaces.

WHY THIS IS USEFUL

Supports a balanced workplace mix that recognises different activities and inclusive choices rather than only peak desk counts.

SP-066

WORKPLACE PEAK DISTRIBUTION

OUTPUT: Peak distribution

EVIDENCE: Aggregate counts + records

SPACE-PLANNING TEAMS QUESTION

Which workplace peaks and troughs are stable enough to plan for, and how do they vary across weekdays, term phases, projects and exceptional periods?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous zone presence, doorway counts and use-duration bands; optional shared-room booking and content-free service indicators. Spatial coverage: defined workplace zones and shared rooms with complete relevant entrances; no individual device or badge history. Sampling and aggregation: one-minute observations aggregated to 15- minute bands across at least eight representative weeks and seasonal samples. Quality gate: commission against manual spot counts; cover every eligible entrance or stated zone; align clocks within 60 seconds; report error, occlusion, saturation, uptime, gaps, resets, drift, relocation, firmware and processing lineage. Counts remain anonymous aggregate observations. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: workplace policy, space/local unit, room-booking, calendar/event, opening, facilities and sensor-data owners. Records: approved operating assumptions, aggregate staffing ranges, events, closures, holidays, planned changes and current space types. Time basis: the selected baseline and future planning horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current timetable, programme, research, workplace, event and room-owner records govern; aggregate observations cannot judge attendance, performance or activity value. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation, comparison and estimate. Technique: distribution and peak-recurrence comparison. Decision interval: the selected representative period. Minimum evidence: valid aggregate coverage plus authoritative calendar and operating context. Method: calculate daily and hourly distributions, recurrence of upper bands, duration and sensitivity to excluded events; separate normal peaks from exceptional demand. Uncertainty: hybrid-policy shifts, incomplete seasonal data, unrecorded visitors and zone spillover. Safe failure: state where peak evidence is too sparse and avoid extrapolating one busy day to the year; otherwise return not assessable.

ANSWER BOUNDARY

Do not identify who attends, infer work habits or performance, compare named teams, or use low presence to withdraw space automatically.

WHY THIS IS USEFUL

Helps planners size shared workplace provision for credible peaks rather than averages or anecdotes.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

7. Teaching, research, workplace and event demand

SP-067 to SP-068 | 10 questions | Translate teaching, research, workplace and event requirements into time-specific demand without judging people or work.

Questions 7-8 of 10

Category 7 of 8

SP-067

EVENT CAPACITY AND DISPLACEMENT

OUTPUT: Event feasibility set

EVIDENCE: Bookings + scenario

SPACE-PLANNING TEAMS QUESTION

Which event formats and audience ranges can the portfolio support without displacing protected teaching, research or workplace commitments?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional anonymous historical event counts and route/shared-facility demand may calibrate ranges; attendee identity or ticket content is not needed. Spatial coverage: event spaces, linked support rooms, public routes and shared facilities with current eligible coverage. Sampling and aggregation: event- and 15-minute aggregates across comparable past events plus the proposed event window. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: events, space, timetable/booking, teaching/research/workplace, capacity, accessibility, facilities, security and safety owners. Records: approved event requirements, current core commitments, room and support-space eligibility, setup/recovery, access, routes, services, closures and event priority rules. Time basis: the proposed event programme and surrounding operating periods. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current timetable, programme, research, workplace, event and room-owner records govern; aggregate observations cannot judge attendance, performance or activity value. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: estimate, simulation and comparison. Technique: constrained capacity and displacement scenario. Decision interval: the proposed event plus setup and recovery. Minimum evidence: owner-approved event requirements, current core commitments and eligible supply. Method: allocate demand only to eligible room classes, calculate displaced or constrained core activity, shared-service peaks and unresolved cases across scenarios. Uncertainty: attendance range, late programme changes, linked-room needs and concurrent events. Safe failure: report unmet demand rather than assuming core activity can move or constraints can be relaxed; otherwise return not assessable.

ANSWER BOUNDARY

Do not approve or book events, displace core activity, predict named attendance, certify crowd safety, or control public access and routes.

WHY THIS IS USEFUL

Shows the true portfolio cost of event scenarios, including linked rooms, setup time and essential activity protection.

SP-068

EVENT SUPPORT AND TURNAROUND

OUTPUT: Event requirement pack

EVIDENCE: Event + service records

SPACE-PLANNING TEAMS QUESTION

What support rooms, routes, amenities, storage, setup, reset time and service-team capacity are required for each proposed event format?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: content-free AV/network/power status, door/lift state and optional anonymous route/amenity demand context. Spatial coverage: the event venue, linked support rooms, setup/loading route, public route, amenities and directly serving assets. Sampling and aggregation: current and proposed event windows, including setup, opening, breaks and recovery. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: event organiser, space/booking, facilities, AV/IT, catering, cleaning, security/access, accessibility and safety owners. Records: event format and requirements, room/linkage rules, setup, loading, amenities, staffing/service availability, access, opening, restrictions, permits and contingency arrangements. Time basis: the full event lifecycle. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current timetable, programme, research, workplace, event and room-owner records govern; aggregate observations cannot judge attendance, performance or activity value. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup and comparison. Technique: event dependency mapping and completeness classification. Decision interval: the full event lifecycle. Minimum evidence: owner-approved event requirement set and current evidence for each linked space/service. Method: map every requirement to a venue, route, asset or service owner; classify confirmed, conditional, missing or conflicting and expose timing dependencies. Uncertainty: late programme changes, vendor details, temporary equipment and status changes. Safe failure: return a dependency/gap map and defer operational approval to the event and service owners; otherwise return not assessable.

ANSWER BOUNDARY

Do not inspect event content, grant access, promise services or staffing, approve permits, operate equipment or certify event suitability or safety.

WHY THIS IS USEFUL

Prevents a venue-only assessment from overlooking support rooms, routes, turnaround and shared-service requirements.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

7. Teaching, research, workplace and event demand

Questions 9-10 of 10

Category 7 of 8

SP-069 to SP-070 | 10 questions | Translate teaching, research, workplace and event requirements into time-specific demand without judging people or work.

SP-069

TIME-SHARING FEASIBILITY

OUTPUT: Time-sharing options

EVIDENCE: Schedule + requirements

SPACE-PLANNING TEAMS QUESTION

Which rooms can support time-separated teaching, research, workplace or event uses without breaching service, access, reset, security or storage constraints?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: content-free equipment/service status and optional anonymous historical use by room; no classification of actual activity from sensing. Spatial coverage: candidate rooms, setup/storage areas, routes and directly serving assets. Sampling and aggregation: session-level availability with setup, recovery and service windows across the planning cycle. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space and room owners, teaching/research/workplace/event owners, booking/timetable, capacity, accessibility, facilities, IT/AV, security and safety owners. Records: permitted use modes, activity requirements, layouts, equipment, storage, cleaning/setup, access, booking rules, service hours and conflicts. Time basis: the proposed shared-use schedule. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current timetable, programme, research, workplace, event and room-owner records govern; aggregate observations cannot judge attendance, performance or activity value. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, simulation and comparison. Technique: multi-mode temporal feasibility and schedule comparison. Decision interval: the proposed shared-use schedule. Minimum evidence: current permitted modes, hard requirements, transition times and eligible booking/service windows. Method: test each use and transition in sequence, preserve setup/storage and inclusive constraints, identify feasible combinations and unresolved handovers. Uncertainty: mode changes requiring approval, setup-duration variation, contamination/confidentiality constraints and asset reliability. Safe failure: exclude any sequence with an unverified mode, transition or service dependency; otherwise return not assessable.

ANSWER BOUNDARY

Do not infer current activity from sensor patterns, change room purpose, publish bookings, or approve shared use, cleaning, access or handover.

WHY THIS IS USEFUL

Shows where flexibility is real and where apparent spare time is blocked by transition, service or governance constraints.

SP-070

FUTURE DEMAND RANGE

OUTPUT: Scenario demand range

EVIDENCE: Forecast scenario

SPACE-PLANNING TEAMS QUESTION

Under each approved enrolment, staffing, research and event scenario, where do shortages or apparent surpluses emerge by space class, location and time band?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: quality-controlled historical aggregate use may calibrate uncertainty only; it cannot replace approved future scenarios. Spatial coverage: the complete portfolio by current room class, with equivalent definitions and explicit unobserved supply. Sampling and aggregation: term-, year- or phase-level historical summaries aligned to the future scenario horizon. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: institutional planning, teaching, research, workplace, events, space, finance, facilities and accessibility owners. Records: approved lower/central/upper demand assumptions, current supply and eligibility, planned additions/losses, capacity/layout, services, closures and policy constraints. Time basis: the stated strategic planning horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current timetable, programme, research, workplace, event and room-owner records govern; aggregate observations cannot judge attendance, performance or activity value. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: calculation, forecast and comparison. Technique: multi-source scenario forecast and threshold analysis. Decision interval: the stated strategic horizon. Minimum evidence: approved demand ranges, current eligible supply and committed portfolio changes. Method: calculate demand and supply by space class and time, identify first constraint crossings, show uncertainty bands and sensitivity to assumptions. Uncertainty: future policy, project success, behavioural adaptation, delayed capital changes and category definitions. Safe failure: report conditional shortage/surplus ranges and never treat apparent surplus as allocation authority; otherwise return not assessable.

ANSWER BOUNDARY

Do not predict named enrolment, staffing or projects, decide priorities, withdraw space, approve capital action or present one scenario as certain.

WHY THIS IS USEFUL

Gives leaders a transparent view of where future pressure is robust and where it depends on uncertain assumptions.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

8. Portfolio performance, flexibility and sustainability

SP-071 to SP-072 | 10 questions | Test portfolio balance, adaptability, resilience, interventions and whole-life sustainability with explicit trade-offs.

Questions 1-2 of 10

Category 8 of 8

SP-071

USE AND SERVICE BURDEN

OUTPUT: Assessment shortlist

EVIDENCE: Use + service evidence

SPACE-PLANNING TEAMS QUESTION

Which spaces combine low supported use with high energy, maintenance or service burden after critical functions, shared loads and resilience value are accounted for?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous room-level use totals, submetered energy or served-zone demand, equipment runtime and content-free asset status where correctly mapped. Spatial coverage: the named rooms, meters or served zones and directly attributable assets; shared loads remain allocated only by an owner-approved method. Sampling and aggregation: 15-minute use and energy summaries over at least one representative annual cycle where available, with maintenance events retained. Quality gate: verify meter-to-room, zone or asset mapping and units; align clocks to the aggregation interval; record calibration or validation, estimated reads, uptime, gaps, resets, drift, shared-load allocation and meter or processing lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, energy, facilities/maintenance, finance, sustainability, room and specialist-function owners. Records: current purpose and criticality, eligible hours, bookings, closures, energy tariffs/emissions factors, maintenance/service records, planned changes and protected functions. Time basis: the agreed representative period and planning horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current portfolio, finance, asset, energy, carbon, project and strategy owners govern their measures and decisions; analytics do not set institutional priorities. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: observation, calculation and comparison. Technique: stratified use-burden comparison. Decision interval: the representative period. Minimum evidence: valid comparable use and service/energy evidence plus current purpose, criticality and room-class context. Method: calculate separate utilisation, energy intensity and service-burden measures; compare only like-for-like classes and identify shared-load uncertainty. Uncertainty: weather, specialist standby loads, resilience roles, deferred maintenance and partial metering. Safe failure: report a multi-dimensional evidence profile and keep spaces with unallocatable shared loads unranked; otherwise return not assessable.

ANSWER BOUNDARY

Do not call a critical or specialist space wasteful, infer occupant behaviour, promise savings, withdraw space, or present allocated energy as a direct room measurement.

WHY THIS IS USEFUL

Identifies candidates for deeper technical and service assessment without treating low use alone as a closure case.

SP-072

ADAPTABILITY AND FLEXIBILITY

OUTPUT: Adaptability profile

EVIDENCE: Survey + scenario

SPACE-PLANNING TEAMS QUESTION

Which spaces can switch among approved uses with the least work, downtime and loss of specialist, inclusive or resilience capability?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: current measured geometry, furniture/equipment inventories, service-point metadata and optional content-free asset status. Spatial coverage: the candidate room, routes, storage and directly serving services for each allowed use mode. Sampling and aggregation: a dated current snapshot and future scenario versions. Quality gate: retain survey method, instrument calibration or validation, survey date, coordinate reference, tolerance, full boundary or route coverage, obscured areas, revision and competent provenance. A survey is a dated observation, not approval. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, room/activity, capacity, accessibility, fire-safety, facilities, M&E, IT/AV, furniture and security owners. Records: permitted uses, approved layouts, geometry, services, equipment, storage, setup/recovery, access, restrictions, warranties and planned changes. Time basis: the stated flexibility horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current portfolio, finance, asset, energy, carbon, project and strategy owners govern their measures and decisions; analytics do not set institutional priorities. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, calculation and comparison. Technique: constraint-aware flexibility scoring and scenario comparison. Decision interval: the stated planning horizon. Minimum evidence: current evidence for all hard constraints and a defined set of allowed use modes. Method: count feasible modes, transition effort, enabling dependencies and capabilities lost; report components rather than one opaque score. Uncertainty: future standards, hidden local requirements, move/setup duration and unverified service capacity. Safe failure: classify adaptability as unknown when any mode depends on missing current evidence; otherwise return not assessable.

ANSWER BOUNDARY

Do not change room purpose, approve reconfiguration, equate flexibility with generic open space, or remove specialist and inclusive provision to improve a score.

WHY THIS IS USEFUL

Shows where genuine reversible flexibility exists and where adaptation would require significant technical or governance action.

8. Portfolio performance, flexibility and sustainability

SP-073 to SP-074 | 10 questions | Test portfolio balance, adaptability, resilience, interventions and whole-life sustainability with explicit trade-offs.

Questions 3-4 of 10

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SP-073

LOW-DEMAND CONSOLIDATION TEST

OUTPUT: Bounded consolidation test

EVIDENCE: Governed scenario

SPACE-PLANNING TEAMS QUESTION

During defined low-demand periods, can eligible activities operate from fewer zones while preserving access, resilience, critical functions and service quality?

SENSORS AND TELEMETRY

Proposed modality and location: anonymous aggregate room/zone use, route totals and energy/service demand; content-free room and asset status. Spatial coverage: all spaces considered for temporary operation, retained alternatives, routes and shared-service zones. Sampling and aggregation: 15-minute representative profiles for low-demand periods and scenario operating windows. Quality gate: verify meter-to-room, zone or asset mapping and units; align clocks to the aggregation interval; record calibration or validation, estimated reads, uptime, gaps, resets, drift, shared-load allocation and meter or processing lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, activity, accessibility, security, facilities, energy, room-booking, IT/AV, safety and service owners. Records: eligible hours, essential/specialist functions, minimum service coverage, access and route constraints, resilience requirements, bookings, closures and operating costs. Time basis: the proposed low-demand periods. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current portfolio, finance, asset, energy, carbon, project and strategy owners govern their measures and decisions; analytics do not set institutional priorities. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: simulation, optimisation and recommendation. Technique: constrained consolidation optimisation and resilience test. Decision interval: the proposed low-demand periods. Minimum evidence: owner-approved essential functions and service constraints, current eligible supply and valid demand profiles. Method: find feasible zone sets that meet all hard constraints, compare energy/service ranges and test failure of one retained room or route. Uncertainty: demand spikes, people needing distributed options, shared-system savings uncertainty and future events. Safe failure: return no consolidation option when inclusive choice, resilience or specialist need cannot be preserved; otherwise return not assessable.

ANSWER BOUNDARY

Do not close floors, restrict access, move activities, promise savings, certify service adequacy or operate building systems.

WHY THIS IS USEFUL

Lets owners examine reversible operating scenarios while making service, inclusion and resilience trade-offs explicit.

SP-074

INTERVENTION SHORTLIST

OUTPUT: Prioritised intervention set

EVIDENCE: Multi-criteria scenario

SPACE-PLANNING TEAMS QUESTION

Which feasible no-cost, operational, reconfiguration, refurbishment or acquisition interventions address the most important portfolio constraints within budget and disruption limits?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: current condition, service and aggregate demand evidence already authorised for the candidate constraints; no extra monitoring is assumed. Spatial coverage: each candidate intervention, affected rooms/routes/services and valid comparator scope. Sampling and aggregation: baseline evidence matched to the proposed delivery horizon and expected benefit interval. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space decision owner, finance, capital projects, facilities, accessibility, sustainability, activity, security and technical owners. Records: approved objectives, candidate interventions, costs/ranges, demand constraints, benefits, risks, dependencies, delivery windows, carbon and lifecycle assumptions. Time basis: the stated planning and budget horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current portfolio, finance, asset, energy, carbon, project and strategy owners govern their measures and decisions; analytics do not set institutional priorities. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: estimate, comparison and recommendation. Technique: constraint reduction and multi-criteria prioritisation. Decision interval: the stated capital or operational horizon. Minimum evidence: owner-approved objectives/weights or a non-weighted trade-off method, current constraints and credible cost/benefit ranges. Method: estimate which constraints each intervention addresses, compare options on separate criteria, show Pareto alternatives and sensitivity to uncertain inputs. Uncertainty: optimism bias, interacting projects, benefits that require consultation and uncertain costs. Safe failure: present conditional option sets and unresolved assumptions, not an automatic investment ranking; otherwise return not assessable.

ANSWER BOUNDARY

Do not approve expenditure, procurement, design or works, assign hidden values, promise benefits, or treat modelled benefit as certified outcome.

WHY THIS IS USEFUL

Gives accountable owners a transparent basis for deciding where detailed business cases are most worthwhile.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

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8. Portfolio performance, flexibility and sustainability

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SP-075 to SP-076 | 10 questions | Test portfolio balance, adaptability, resilience, interventions and whole-life sustainability with explicit trade-offs.

SP-075

PORTFOLIO MIX BALANCE

OUTPUT: Supply-demand gap map

EVIDENCE: Portfolio records + forecast

SPACE-PLANNING TEAMS QUESTION

Does the current mix of room types, sizes, service levels and locations align with supported demand, fallback capacity and strategic commitments?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: optional quality-controlled anonymous historical use by verified room class, kept separate from scheduled and requested demand. Spatial coverage: the complete portfolio scope and each current room class, including unobserved or restricted supply. Sampling and aggregation: term-, year- and peak-period summaries matched to the strategic horizon. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: institutional planning, space, teaching, research, workplace, events, accessibility, facilities and sustainability owners. Records: current supply by room class, capacity/layout, permitted uses, demand scenarios, service and resilience requirements, planned additions/losses and closures. Time basis: the current baseline and approved future horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current portfolio, finance, asset, energy, carbon, project and strategy owners govern their measures and decisions; analytics do not set institutional priorities. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: calculation, forecast and comparison. Technique: multi-class supply-demand and resilience comparison. Decision interval: the stated planning horizon. Minimum evidence: current eligible supply, owner-approved demand ranges and minimum resilience constraints. Method: compare capacity and time availability by class, identify robust and scenario-dependent gaps, and test whether apparent excess can substitute across classes. Uncertainty: category definitions, uncertain future demand, specialist/non-substitutable rooms and partial observed-use coverage. Safe failure: report imbalance ranges without labelling rooms surplus or recommending withdrawal; otherwise return not assessable.

ANSWER BOUNDARY

Do not collapse distinct room types into one area total, decide portfolio targets, withdraw provision, or equate low observed use with excess supply.

WHY THIS IS USEFUL

Shows whether the building has the right kinds of spaces, not merely enough total floor area.

SP-076

PORTFOLIO RESILIENCE STRESS TEST

OUTPUT: Stress-test results

EVIDENCE: Governed scenario

SPACE-PLANNING TEAMS QUESTION

How well can the portfolio absorb temporary loss of a key room, floor, route, utility or specialist service without breaching hard requirements?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: owner-issued room, route and service status plus optional historical aggregate demand for calibration. Spatial coverage: the failed component, affected spaces, eligible alternatives, routes and shared-service dependencies. Sampling and aggregation: the stated outage durations and matched demand periods. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space, activity, accessibility, facilities, security, fire/emergency, IT/AV, M&E; and room-booking owners. Records: current inventory and demand, hard requirements, alternative spaces/routes, service dependencies, contingency arrangements, planned outages and concurrent risks. Time basis: the stated outage scenarios and planning horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current portfolio, finance, asset, energy, carbon, project and strategy owners govern their measures and decisions; analytics do not set institutional priorities. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: simulation and estimate. Technique: portfolio stress-test and constrained reallocation simulation. Decision interval: the stated outage scenarios. Minimum evidence: current topology, eligible alternatives and owner-approved demand/hard constraints. Method: remove each component, re-test feasible capacity and routes without relaxing constraints, calculate unmet demand and identify cascading single points. Uncertainty: compound failures, future status, unmodelled informal workarounds and demand adaptation. Safe failure: show conditional resilience and unresolved cases; do not declare operational continuity; otherwise return not assessable.

ANSWER BOUNDARY

Do not declare or manage an outage, assign alternatives, relax safety/accessibility, certify continuity or command emergency action.

WHY THIS IS USEFUL

Reveals where flexibility is only apparent because several spaces depend on the same fragile route or service.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

8. Portfolio performance, flexibility and sustainability

SP-077 to SP-078 | 10 questions | Test portfolio balance, adaptability, resilience, interventions and whole-life sustainability with explicit trade-offs.

Questions 7-8 of 10

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SP-077

WHOLE-LIFE CARBON OF OPTIONS

OUTPUT: Carbon range comparison

EVIDENCE: Metering + lifecycle data

SPACE-PLANNING TEAMS QUESTION

What are the bounded operational and embodied-carbon consequences of retaining, reconfiguring, consolidating or relocating the current arrangement?

SENSORS AND TELEMETRY

Proposed modality and location: submetered energy or served-zone demand, equipment runtime and environmental context where correctly mapped; no occupant-level energy attribution. Spatial coverage: affected rooms, meters/zones, assets and intervention quantities, with shared loads separately modelled. Sampling and aggregation: 15-minute operational data over a representative annual baseline; embodied quantities tied to the scenario delivery dates. Quality gate: verify meter-to-room, zone or asset mapping and units; align clocks to the aggregation interval; record calibration or validation, estimated reads, uptime, gaps, resets, drift, shared-load allocation and meter or processing lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: energy/carbon, sustainability, space, facilities, project, procurement, finance and asset owners. Records: current energy/emissions data, scenario schedules, material/equipment quantities, product or lifecycle data, waste/reuse assumptions, service-life horizon and approved carbon factors. Time basis: the scenario construction and operating horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current portfolio, finance, asset, energy, carbon, project and strategy owners govern their measures and decisions; analytics do not set institutional priorities. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: calculation, estimate and comparison. Technique: bounded lifecycle-carbon calculation and scenario comparison. Decision interval: the stated delivery and operating horizon. Minimum evidence: credible operational baseline, scenario quantities, service-life assumptions and current owner-approved factors. Method: calculate operational and embodied ranges separately, include reuse and displacement effects, test sensitivity to utilisation and service-life assumptions and avoid double counting. Uncertainty: shared meters, future grid factors, product data quality, rebound effects and uncertain retained-asset life. Safe failure: show ranges and evidence grades; withhold totals when material quantities or energy attribution are not defensible; otherwise return not assessable.

ANSWER BOUNDARY

Do not present estimates as formally validated carbon performance, promise savings, approve a project, or attribute energy behaviour to individuals or teams.

WHY THIS IS USEFUL

Lets planners compare whether physical change is justified once both operational and embodied consequences are visible.

SP-078

EXPLICIT PORTFOLIO TRADE-OFFS

OUTPUT: Trade-off frontier

EVIDENCE: Multi-criteria scenario

SPACE-PLANNING TEAMS QUESTION

How do feasible options compare across capacity, utilisation, inclusion, resilience, cost, disruption, adaptability and carbon when no option dominates?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: only the quality-controlled aggregate evidence already justified for each criterion; no single sensor can represent overall performance. Spatial coverage: the same spatial and temporal scope for every compared option, with criterion-specific coverage declared. Sampling and aggregation: baseline and scenario horizons aligned to decision dates and service life. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space decision owner, finance, sustainability, accessibility, facilities, activity, security, project and technical owners. Records: feasible options, demand, capacity, access, resilience, costs, carbon, disruption, dependencies, consultation evidence and owner-defined hard constraints. Time basis: the strategic decision horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current portfolio, finance, asset, energy, carbon, project and strategy owners govern their measures and decisions; analytics do not set institutional priorities. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: comparison and recommendation. Technique: multi-criteria Pareto comparison and sensitivity analysis. Decision interval: the strategic decision horizon. Minimum evidence: only options that pass all hard constraints plus transparent criterion definitions and owner-approved weights if any. Method: show non-dominated options, criterion ranges, key trade-offs and sensitivity to weights/assumptions; keep qualitative consultation findings distinct. Uncertainty: incommensurable values, uncertain future demand, incomplete costs/carbon and stakeholder disagreement. Safe failure: avoid a single score when it would conceal conflict; return an option set and unresolved choices; otherwise return not assessable.

ANSWER BOUNDARY

Do not set institutional values, invent weights, choose the final option, certify sustainability or inclusion, or present one modelled ranking as objective truth.

WHY THIS IS USEFUL

Supports accountable deliberation by showing where each feasible option gains and loses rather than masking trade-offs.

8. Portfolio performance, flexibility and sustainability

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SP-079 to SP-080 | 10 questions | Test portfolio balance, adaptability, resilience, interventions and whole-life sustainability with explicit trade-offs.

SP-079

PORTFOLIO INDICATOR ASSURANCE

OUTPUT: Assured indicator set

EVIDENCE: Provenance metadata

SPACE-PLANNING TEAMS QUESTION

Which portfolio indicators are reliable enough to report now, and what coverage, age, uncertainty, exclusions and conflict statement must accompany each?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: quality and provenance metadata for anonymous occupancy, environmental, energy and asset evidence used by each indicator. Spatial coverage: the exact rooms, zones, meters, assets and periods represented by the indicator. Sampling and aggregation: the reporting interval with native observation times and declared aggregation. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: space information, booking/timetable, facilities, energy, finance, accessibility, sustainability and sensor-data owners. Records: current inventory, capacities, eligible hours, use, costs, energy/carbon, condition, flexibility and inclusion evidence with official definitions and versions. Time basis: the stated reporting date and comparator periods. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current portfolio, finance, asset, energy, carbon, project and strategy owners govern their measures and decisions; analytics do not set institutional priorities. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: calculation and comparison. Technique: indicator calculation and evidence-strength classification. Decision interval: the stated reporting period. Minimum evidence: a published definition, authoritative denominator and claim-specific valid evidence. Method: calculate each indicator separately, expose coverage/exclusions, classify direct, allocated, estimated or unavailable evidence and compare source versions. Uncertainty: changing definitions, partial monitoring, shared loads, stale inventory and small sensitive groups. Safe failure: publish only supported indicators, suppress unsafe detail and label the rest not assessable; otherwise return not assessable.

ANSWER BOUNDARY

Do not present an estimate as officially validated fact, combine incompatible metrics, rank people or units, or use a dashboard indicator to authorise space change.

WHY THIS IS USEFUL

Produces transparent portfolio reporting that decision-makers can interpret without mistaking completeness of presentation for completeness of evidence.

SP-080

DECISION-READY PORTFOLIO PACK

OUTPUT: Decision-readiness map

EVIDENCE: Evidence + authority

SPACE-PLANNING TEAMS QUESTION

Which candidate actions are ready for accountable consideration, which require named further work or consultation, and which are currently not assessable?

SENSORS AND TELEMETRY

No physical sensing is required for the core answer. Proposed corroborating integration or telemetry, if permissioned: source-health, mapping and quality metadata for every proposed telemetry source; no new sensing is assumed. Spatial coverage: each candidate action, affected spaces/routes/services and every evidence dependency. Sampling and aggregation: current at the stated decision checkpoint and future-effective dates within the action horizon. Quality gate: calibration is not applicable to record feeds; validate identifiers, schema and state semantics, effective, event and retrieval times, clock handling, expected refresh, completeness, uptime, gaps, stale-state timeout, change history and source lineage. Use only after permission, exact mapping, commissioning or validation and current health are independently verified; otherwise mark the evidence unavailable.

AUTHORITATIVE SUPPORTING DATA

Owners: the accountable institutional space owner and all activity, capacity, accessibility, safety, facilities, security, finance, sustainability, project and information owners touched by each action. Records: candidate actions, current inventory, demand, constraints, surveys, consultation, costs, carbon, risks, dependencies, approvals and unresolved conflicts. Time basis: the stated decision checkpoint and action horizon. Time and provenance: keep effective, event or observation and retrieval times separate; preserve owner, record ID, version, permission and transformations. Current portfolio, finance, asset, energy, carbon, project and strategy owners govern their measures and decisions; analytics do not set institutional priorities. Apply owner staleness rules or show age. Keep equal-authority conflicts visible and block high-impact use. The May 2021 floor plan is historical spatial context only.

ANALYSIS REQUIRED

Operations: lookup, comparison and recommendation. Technique: evidence-readiness and authority classification. Decision interval: the stated decision checkpoint. Minimum evidence: a complete requirement matrix for each action and current source/owner status. Method: classify every criterion as confirmed, conditional, missing, conflicting, expired, permission-restricted or outside competence; separate evidence readiness from decision authority. Uncertainty: new constraints, changing demand, unresolved stakeholder values and evidence that may expire before action. Safe failure: provide a supported action/evidence map, named next owner and explicit not-assessable cases; never output an approval; otherwise return not assessable.

ANSWER BOUNDARY

Do not approve, reject, allocate, withdraw, fund or schedule an action, conceal unresolved conflicts, or imply that analytical completeness replaces accountable judgement and consultation.

WHY THIS IS USEFUL

Creates a decision pack with evidence-ready options, named next work, unresolved conflicts and honest abstention.

Universal boundary: no identity or movement tracking; no productivity, performance or protected-trait inference; no allocation, certification, authorisation or system control.

Portfolio coverage and decision map

Every category contains ten distinct planning decisions; the catalogue is a route to evidence and accountable action, not a scorecard.

CATEGORY	DECISION COVERAGE	PRIMARY EVIDENCE AND HUMAN INPUT	TYPICAL OUTPUTS
1 Space inventory, ownership and current use	Current identity, rights, purpose, availability, area and service dependencies.	Space and Estates records; local owners; controlled drawings; measured survey.	Baseline, crosswalk, rights matrix, gap list.
2 Occupancy, utilisation and evidence quality	Representative use, peak pressure, low use, booking mismatch, context and evidence fitness.	Anonymous aggregate counts; booking/opening records; source-health metadata; proportionate sampling.	Profiles, mismatch queue, metric specification.
3 Capacity, layouts, equipment and suitability	Approved capacity, requirement fit, installed state, layout options, blockers and change-of-use evidence.	Capacity/layout/asset owners; measured surveys; service telemetry; competent technical input.	Eligible rooms, constraint register, due-diligence pack.
4 Adjacencies, circulation and shared facilities	Mandatory adjacencies, shared-facility demand, circulation, route conflicts, failure dependencies and location options.	Owner requirements; route/facility records; count-only flow evidence; consultation; governed scenarios.	Dependency maps, hotspot schedules, option comparison.
5 Accessibility, inclusion and diverse user needs	Accessible journeys, stated needs, layout barriers, provision resilience, sensory choice and consultation gaps.	Access audits; person-stated requirements; current status; measured/environmental surveys; consultation.	Evidence map, consultation brief, qualified option set.
6 Moves, scenarios, decants and change impacts	Move impacts, destination and decant gates, sequencing, temporary loss, enabling work, consultation and reversibility.	Project/change records; owner decisions; surveys; bookings; aggregate before/after evidence.	Impact register, sequence, blocker list, learning plan.
7 Teaching, research, workplace and event demand	Teaching, research, workplace and event demand, dependencies, time-sharing and future ranges.	Timetable/programme/event records; technical consultation; anonymous aggregate calibration; approved scenarios.	Demand envelopes, mismatch queues, feasibility sets.
8 Portfolio performance, flexibility and sustainability	Use and service burden, adaptability, consolidation, interventions, mix, resilience, carbon and reporting.	Portfolio/finance/asset/energy/carbon records; lifecycle data; governed multi-criteria scenarios.	Stress tests, trade-off frontier, assured indicators.

Coverage commitments

CURRENT STATE

Inventory, rights, purpose, condition, capacity and evidence quality are separated from scenarios.

CHANGE AND DISRUPTION

Moves, decants, closures, consultation, sequencing, fallback and learning are explicit.

INCLUSION AND CHOICE

Stated requirements, accessible journeys, diverse options and competent assessment are preserved.

LONG-TERM VALUE

Flexibility, resilience, service burden, portfolio mix, cost and whole-life carbon remain visible together.

Evidence, permissions and decision-handover rules

Apply these rules before any answer is placed in a planning brief, consultation pack, option appraisal or governance paper.

1. START WITH THE DECISION

Name the planning choice, affected spaces and decision date before collecting data.

2. USE THE CURRENT OWNER RECORD

Retrieve the source valid for that date. Record owner, identifier, version, permission and retrieval time.

3. MAP THE EXACT CLAIM

Verify the room, route, asset, meter, entrance and service boundary. Shared or adjacent evidence stays contextual.

4. VERIFY PROPOSED EVIDENCE

Confirm modality, placement, coverage, aggregation, calibration or validation, clock, uptime, gaps, drift and lineage.

5. MINIMISE DATA

Use anonymous aggregates, numerical conditions and the least intrusive method. Do not collect content or identity trails.

6. SEPARATE OPERATIONS

Label lookup, observation, calculation, estimate, forecast, comparison, diagnosis and recommendation distinctly.

7. STATE MINIMUM EVIDENCE

Define the evidence threshold in advance. Missing hard constraints trigger qualification, owner deferral or not assessable.

8. KEEP CONFLICTS VISIBLE

Never average or silently choose between equal-authority records. Name the owners needed to resolve the conflict.

9. INCLUDE PEOPLE AND PROFESSIONALS

Use accessible consultation and person-stated requirements; obtain competent accessibility, fire, safety and technical input.

10. HAND OVER AUTHORITY

The final answer names evidence, uncertainty, next owner and decision boundary. It never allocates, certifies, authorises or controls.

Decision-handover check

Decision and date named

Exact scope mapped

Current owners cited

Permissions valid

Minimum evidence met

Uncertainty stated

Consultation recorded

Authority remains with owner